

SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The center and the six vertices canonically label the seven covering triangles. A finite classification routes every hypothetical cover to one of three strategies: trace-length deficit, area loss, or a finite equilateral-enclosure obstruction. The finite-enclosure strategy includes the explicit zero-gap nine-point theorem and the nonzero-gap residual witness sets. Exact local reach calculations are retained only as certificates that the relevant finite sets do not fit in a unit C triangle. The paper is organized as one proof, and the two mixed support-arc intersections are verified in a self-contained exact certificate appendix over $\mathbb{Q}(\sqrt{3})$.

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1. INTRODUCTION AND PROOF OUTLINE

1.1. The theorem and the canonical labeling. Let

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right), \quad H = \text{conv}\{V_0, \dots, V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \quad r_i = [O, V_i], \quad M_i = \frac{1}{2}V_i,$$

and put

$$S = \partial H \cup \bigcup_{i=0}^5 r_i.$$

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Then

$$\mathcal{H}^1(\partial H) = 6, \quad \mathcal{H}^1\left(\bigcup_{i=0}^5 r_i\right) = 6, \quad \mathcal{H}^1(S) = 12.$$

Theorem 1.1 (Main theorem). *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem). *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least one, whereas any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the seven covering triangles admit a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \quad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle* at V_i . Set

$$T_C = \overline{U_C}, \quad T_i = \overline{U_i}.$$

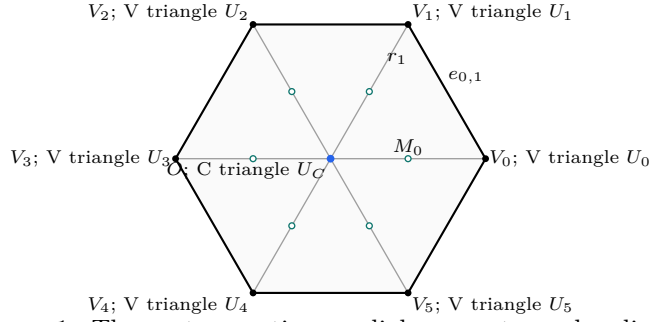
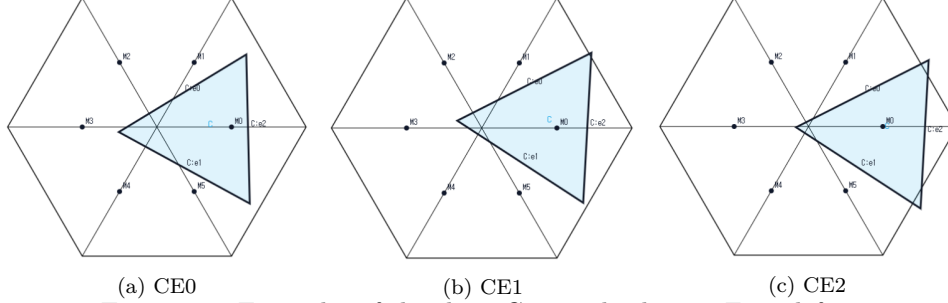


FIGURE 1. The center, vertices, radial segments, and radial midpoints of H .

1.2. Finite classifications. The center type is the number of boundary edges on which T_C has a positive-length trace. The only possibilities are 0, 1, 2, denoted by CE0, CE1, CE2.



(a) CE0 (b) CE1 (c) CE2
 FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of ∂H ; point contacts do not count. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

For the V triangle T_i , let $o(T_i)$ be the number of its triangle vertices outside H and put

$$n(T_i) = |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|,$$

with indices modulo 6. The raw pair $(3, 0)$ says that all three vertices of T_i lie outside H ; it does not say that T_i is disjoint from H , since $V_i \in \text{int}(T_i) \cap H$. Proposition 2.5 replaces every such triangle by a same-orientation translate with exactly the same trace in H , still containing V_i in its interior, and with at most two vertices outside H . We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

type	$(o(T_i), n(T_i))$
Vd0	$(1, 0), (2, 0)$
Vd1	$(1, 1)$
Vd2	$(1, 2)$
T3-like	$(2, 1)$

Let A_i, B_i, C_i be the maximal reaches of T_i toward V_{i-1}, V_{i+1}, O , respectively. The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}|.$$

Lowercase (a_i, b_i, c_i) always denotes selected lower bounds for these reaches and never the maxima defining N_+ .

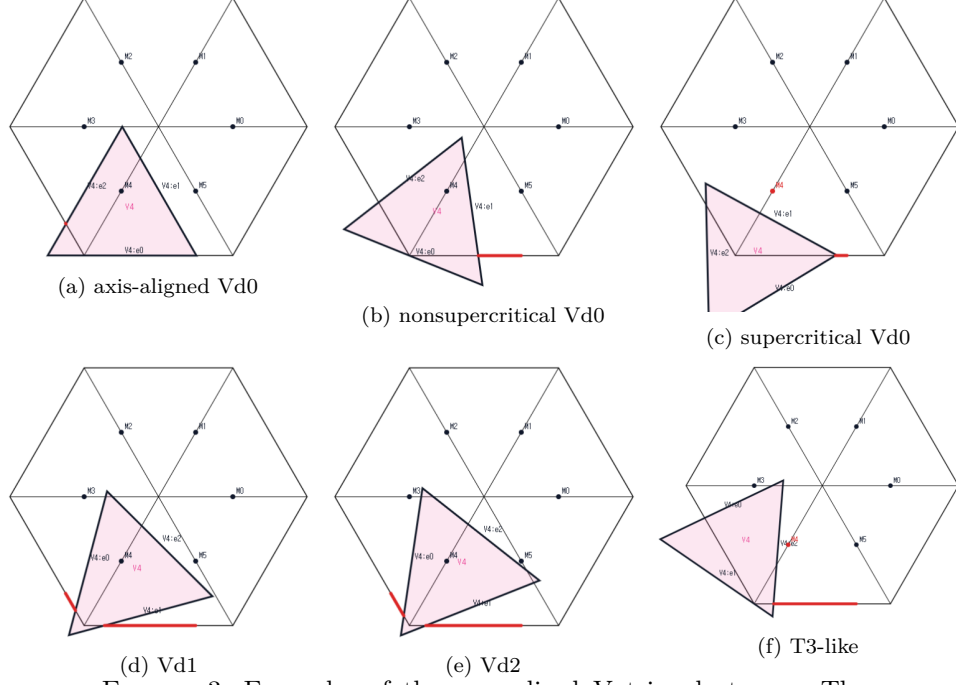


FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \leq 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

On $e_{i,i+1}$, parametrized from V_i to V_{i+1} , the two incident open traces have the form $[0, B_i]$ and $(1 - A_{i+1}, 1]$. Their nonempty complement

$$[B_i, 1 - A_{i+1}]$$

is a *boundary gap*; equality gives a singleton gap, which remains uncovered. Let N_{gap} be the number of positive center traces that contain a boundary gap.

For the case tables, put

$$d = |\{i : T_i \text{ is Vd1 or Vd2}\}|, \quad t = |\{i : T_i \text{ is T3-like}\}|,$$

and let $N_{\text{sp}} = d + t$. The common trace calculation proves that the number of V triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\text{sp}}.$$

In particular, every CE1/CE2 configuration with $N_+ + N_{\text{sp}} \geq 3$ is impossible by a trace-length bound on S .

1.3. Exhaustive routing. Lemma 2.8 proves the fundamental equivalence

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Thus Table 1 first dispatches every zero-gap case without reference to the center class. The nonzero-gap part is read from top to bottom; the first applicable row is

the unique route for a hypothetical cover. The full classification proof is Proposition 2.14.

N_{gap}	N_+	vertex-type refinement	center class	strategy
0	0	all (d, t)	CE0/CE1/CE2	1
0	1	$d \geq 1$	CE0/CE1/CE2	1
0	1	$d = 0, t \geq 1$	CE0/CE1/CE2	2
0	1	$(d, t) = (0, 0)$	CE0/CE1/CE2	3
0	≥ 2	all (d, t)	CE0/CE1/CE2	2
≥ 1	≥ 2	all (d, t)	CE1/CE2	1
≥ 1	0 or 1	$N_+ + d + t \geq 3$	CE1/CE2	1
≥ 1	0	$(d, t) = (0, 0)$	CE1/CE2	3
≥ 1	0	$d \geq 1, d + t \leq 2$	CE1/CE2	1
≥ 1	0	$d = 0, t \in \{1, 2\}$	CE1/CE2	3
≥ 1	1	$(d, t) = (0, 0)$	CE1/CE2	3
≥ 1	1	$(d, t) = (0, 1)$	CE1/CE2	3
≥ 1	1	$(d, t) = (1, 0)$	CE1	1
≥ 1	1	$(d, t) = (1, 0)$	CE2	1+3

TABLE 1. The exhaustive routing. Strategy 1 is a trace-length argument, Strategy 2 sums area loss, and Strategy 3 forces a finite residual set into the C triangle and proves an equilateral-enclosure obstruction. Nonzero gap count forces center class CE1 or CE2. In that part the high-count rows are applied first; the final two rows separate the CE1 length route from the detailed CE2 finite-enclosure placement assembly.

1.4. Organization of the proof. Section 2 proves the structural classification, strict handoff selection, midpoint facts, and the signed center normal form. Section 3 gives the complete trace-length argument. Section 4 proves the local and cyclic area-loss estimates. Section 5 develops the finite residual-hull principle, the exact local reach certificates for the nonzero-gap witnesses, and the explicit zero-gap nine-point obstruction. Section 6 then applies the zero-gap matrix before the nonzero-gap rows of Table 1 and closes the proof.

2. STRUCTURAL REDUCTION AND COMMON GEOMETRY

This section supplies the complete finite reduction used by every later method. It proves the open–closed scaling equivalence, the center and vertex classifications, strict boundary handoffs, midpoint facts, and exhaustive routing. It then records the signed center normal form used in both the trace and finite-enclosure arguments. Boundary reaches are used only inside direct geometric inequalities; no global composition of reach maps is part of the proof. Each result is stated and proved once.

2.1. Structural Reduction to Finite Cases. Throughout, indices are taken in $\mathbb{Z}/6\mathbb{Z}$. Put

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right), \quad H = \text{conv}\{V_0, \dots, V_5\}.$$

Thus H is the regular hexagon of side length one. We use the notation

$$e_{i,i+1} = [V_i, V_{i+1}], \quad r_i = [O, V_i], \quad M_i = \frac{1}{2}V_i,$$

and, when needed,

$$D = \bigcup_{i=0}^5 r_i, \quad S = \partial H \cup D, \quad S_{1/2} = \partial H \cup \{O, M_0, \dots, M_5\}.$$

The arms r_i meet only at O , up to sets of one-dimensional measure zero, and consequently

$$(1) \quad \mathcal{H}^1(\partial H) = 6, \quad \mathcal{H}^1(D) = 6, \quad \mathcal{H}^1(S) = 12.$$

For $L > 0$, $H_L := LH$ denotes the concentric regular hexagon of side length L . All symmetries used below belong to the dihedral group D_6 ; applying a symmetry always transports the indices, orientations, and incoming and outgoing labels together.

2.1.1. Open, closed, and scaled formulations.

Proposition 2.1 (Compactness and scaling equivalence). *The following assertions are equivalent.*

- (1) *The set H is covered by seven open equilateral triangles of side length 1.*
- (2) *For some $\varepsilon > 0$, the set H is covered by seven closed equilateral triangles of side length $1 - \varepsilon$.*
- (3) *For some $L > 1$, the set H_L is covered by seven closed equilateral triangles of side length 1.*

We conduct the geometric argument in the open side-one model. Denote the original open triangles by U_C and U_i for $0 \leq i \leq 5$, and put

$$T_C = \overline{U_C}, \quad T_i = \overline{U_i}.$$

The distinction between U_i and T_i will be maintained: closure may add endpoints or tangencies, although it does not change area or one-dimensional trace measure.

Lemma 2.2 (Distinct triangles). *The seven triangles can be labelled so that*

$$O \in U_C, \quad V_i \in U_i \quad (0 \leq i \leq 5),$$

and these seven triangles are distinct. Consequently

$$O \in \text{int}(T_C), \quad V_i \in \text{int}(T_i).$$

2.1.2. The center classification.

For a closed C triangle define

$$N_E(T_C) = |\{i : T_C \cap e_{i,i+1} \text{ contains a positive-length interval}\}|.$$

Point contacts, endpoint contacts, and tangencies do not contribute. We say that T_C has type CE0, CE1, or CE2 according as $N_E(T_C)$ is 0, 1, or 2.

Proposition 2.3 (Exhaustive center types). *Every C triangle has exactly one of the types CE0, CE1, and CE2. Equivalently,*

$$N_E(T_C) \leq 2.$$

For the CE1/CE2 types, Proposition 2.12 below records the additional fact that T_C contains exactly one radial midpoint.

2.1.3. *Vertex types and exact-trace normalizations.* For the closure T_i of an original open V triangle U_i , let

$$o(T_i) = |\{\text{vertices of } T_i \text{ outside } H\}|$$

and put

$$n(T_i) = |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|.$$

Lemma 2.4 (Local wedge). *Every original V triangle has $1 \leq o \leq 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one of its vertices belongs to H ; if both $\mathcal{H}^1(T_i \cap r_{i-1}) > 0$ and $\mathcal{H}^1(T_i \cap r_{i+1}) > 0$, at least two of its vertices belong to H .*

We use the following orientation normal forms for both translations below. Set

$$D_t = 1 - t + t^2, \quad z = \sqrt{D_t}, \quad 0 \leq t \leq 1.$$

After relabelling the vertices of the equilateral triangle, every orientation belongs to one of the two families

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

In Type I write

$$(2) \quad U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\},$$

and in Type II write

$$(3) \quad U = \alpha + (1 - t)x + ty, \quad V = \beta + tx - y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\}.$$

In either form, $V_0 \in \text{int}(T)$ is equivalent to

$$(4) \quad \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < z.$$

For later use, the Type-II vertex in the wedge is

$$(5) \quad Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right).$$

Both coordinates are positive under (4).

Proposition 2.5 (Exact-trace normalization of raw $(3, 0)$ roles). *Let U be an original open V_i -triangle, put $T = \bar{U}$, and suppose $(o(T), n(T)) = (3, 0)$. There is a translate $\hat{U}$ of U , with closure $\hat{T}$, the same orientation, and the same side length, such that*

$$(6) \quad V_i \in \hat{U}, \quad (o(\hat{T}), n(\hat{T})) = (2, 0),$$

and both the closed and open traces on the hexagon are unchanged:

$$(7) \quad T \cap H = \hat{T} \cap H, \quad U \cap H = \hat{U} \cap H.$$

Consequently the three maximal reaches on the two incident edges and the own radial segment are unchanged. In the notation introduced below, the values A_i, B_i, C_i , the inequality $A_i + B_i > 1$, and hence N_+ are all preserved.

Apply Proposition 2.5 simultaneously to every raw $(3, 0)$ vertex role and then reuse the symbols U_i, T_i for the translated roles. The rolewise equalities (7) preserve the entire open cover, every closed trace, the condition $V_i \in U_i$, all actual reaches, and N_+ . This simultaneous exact-trace normalization is the convention throughout the rest of the paper.

For the normalized roles, we use the names

type	(o, n)
Vd0	$(1, 0), (2, 0)$
Vd1	$(1, 1)$
Vd2	$(1, 2)$
T3-like	$(2, 1)$.

Proposition 2.6 (Exhaustive normalized vertex types). *Every normalized V triangle has exactly one of the four types Vd0, Vd1, Vd2, and T3-like.*

The next normalization is useful for trace estimates, but its final warning is essential.

Proposition 2.7 (T3-like closed-trace domination). *Let T be the closure of an original T3-like V_i -triangle. There is a translate $\widehat{T}$ of T , with the same orientation and side length, such that*

$$V_i \in \text{relint}(a \text{ side of } \widehat{T}), \quad T \cap H \subsetneq \widehat{T} \cap H,$$

and $\widehat{T}$ remains of type $(o, n) = (2, 1)$. Every old open interior point in $H \setminus \{V_i\}$ remains an interior point of $\widehat{T}$. The point V_i itself becomes a boundary point, so $\widehat{T}$ is a closed-trace majorant and is not a replacement open triangle.

2.1.4. *Reach coordinates, supercritical V triangles, and gaps.* For the V triangle U_i under the simultaneous normalization convention, define its maximal incoming, outgoing, and radial reaches by

$$\begin{aligned} A(U_i) &= \sup_{\substack{t \in [0,1] \\ V_i + t(V_{i-1} - V_i) \in U_i}} t, \\ B(U_i) &= \sup_{\substack{t \in [0,1] \\ V_i + t(V_{i+1} - V_i) \in U_i}} t, \\ C(U_i) &= \sup_{\substack{t \in [0,1] \\ V_i + t(O - V_i) \in U_i}} t. \end{aligned}$$

Closure does not change the suprema, so

$$A(U_i) = A(T_i), \quad B(U_i) = B(T_i), \quad C(U_i) = C(T_i).$$

For compact indexed formulas, put

$$A_i := A(U_i) = A(T_i), \quad B_i := B(U_i) = B(T_i), \quad C_i := C(U_i) = C(T_i).$$

The closure T_i contains the corresponding endpoints. We reserve lowercase (a_i, b_i, c_i) for selected or prescribed reach lower bounds whenever actual and selected values occur together. A realizing triangle then satisfies

$$A_i \geq a_i, \quad B_i \geq b_i, \quad C_i \geq c_i.$$

An actual V triangle is supercritical if $A_i + B_i > 1$, and

$$(8) \quad N_+ = |\{i : A_i + B_i > 1\}|.$$

This definition never counts a smaller reach pair in place of the actual V triangle.

Parametrize $e_{i,i+1}$ by

$$X_i(x) = V_i + x(V_{i+1} - V_i), \quad 0 \leq x \leq 1.$$

The two incident open traces are

$$U_i \cap e_{i,i+1} = [0, B_i), \quad U_{i+1} \cap e_{i,i+1} = (1 - A_{i+1}, 1]$$

in this coordinate. A *boundary gap* on this edge is the nonempty closed complement between these two traces,

$$[B_i, 1 - A_{i+1}] \quad \text{when } B_i \leq 1 - A_{i+1}.$$

In particular, equality gives a singleton gap. This convention retains the actual open traces; a point missing from both open triangles is not erased merely because both closures contain it.

Lemma 2.8 (Exhaustiveness of the gap split). *On each edge the complement of the two incident V triangle traces is either empty or one closed interval; a singleton interval counts as a gap. Every edge having a gap has positive-length intersection with the open center triangle. Consequently a CE0 center permits no gaps, a CE1 center permits at most one, and a CE2 center permits at most two. Moreover, under a hypothetical cover,*

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Lemma 2.9 (T3-like V triangles are nonsupercritical). *Every original T3-like V triangle satisfies*

$$(9) \quad A_i + B_i \leq 1.$$

In particular, it is not counted by N_+ .

We state the precise passage from maximal reaches to strict handoff lower bounds. Its hypothesis that the six V triangles themselves cover the boundary is part of the statement.

Proposition 2.10 (Strict boundary handoffs). *Assume that $U_0, \dots, U_5$ cover ∂H . On each edge choose*

$$\xi_i \in (1 - A_{i+1}, B_i)$$

and define

$$a_i = 1 - \xi_{i-1}, \quad b_i = \xi_i.$$

Then $0 < a_i < A_i$, $0 < b_i < B_i$, the two selected anchors lie in U_i , and

$$a_i^2 + a_i b_i + b_i^2 < 1.$$

Moreover:

- (1) *if exactly one actual V triangle, indexed by p , is supercritical, every strict selection has exactly one selected supercritical V triangle, also indexed by p ;*
- (2) *if at least two actual V triangles are supercritical, some simultaneous strict selection has at least two selected supercritical V triangles.*

2.1.5. *Midpoint forcing.* We first need a local fact for V triangles. Take unit vectors

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \quad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2} \right).$$

Lemma 2.11 (Self-midpoint obstruction). *Let a closed unit triangle contain*

$$0, \quad au, \quad bv, \quad \frac{u+v}{2}.$$

Then $a + b \leq 1$. Consequently, for every V triangle,

$$M_i \in T_i \implies A_i + B_i \leq 1.$$

If the two demanded boundary points and the midpoint are contained with a positive margin, the conclusion is strict.

The corresponding center fact is exact and uses the interior condition at O .

Proposition 2.12 (CE1/CE2 exactly-one-midpoint property). *Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$. If T_C is CE1 or CE2, equivalently if it has positive-length intersection with a boundary edge of H , then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

Remark 2.13. The hypothesis $O \in \text{int}(T_C)$ cannot be weakened to $O \in T_C$: the triangle $\text{conv}\{O, V_0, V_1\}$ has a boundary-edge overlap and contains both M_0 and M_1 .

Proposition 2.14 (Exhaustive structural reduction). *Every hypothetical cover by seven open unit equilateral triangles admits the distinct center and V triangles used in Table 1, and its maximal reaches determine exactly one row of that table.*

2.2. Signed CE1/CE2 Reductions. This section gives the signed center model used by both later strategies. Its boundary traces and six radial exits are inserted directly into the finite-witness arguments of Section 5.

2.2.1. One signed center normal form.

Proposition 2.15 (Signed CE1/CE2 center normal form). *Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$\begin{aligned} 0 < R < 1, \quad W &= 1 - R, \\ E &= \sqrt{1 - RW}, \quad \eta = 1 - E, \quad P = E(1 - E), \end{aligned}$$

and nonnegative center slacks α, δ such that, with

$$k = \eta + \alpha + \delta,$$

the C triangle is

$$(10) \quad \boxed{\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ F_1 &= Rb + Wa - k, \\ F_2 &= W + \delta - b + Ra, \end{aligned}} \quad T_C = \{F_0, F_1, F_2 \geq 0\}.$$

Define

$$(11) \quad \Delta_R = P - \alpha - W\delta, \quad \Delta_L = P - R\alpha - \delta.$$

Then

$$(12) \quad T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta \right], \quad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R},$$

and the possible companion trace is

$$(13) \quad T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha \right] \quad \text{when } \Delta_L > 0, \quad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}.$$

The normalization requires $\Delta_R > 0$, and

$$(14) \quad T_C \text{ is CE1} \iff \Delta_L \leq 0, \quad T_C \text{ is CE2} \iff \Delta_L > 0.$$

The six center exits, measured from O , are

$$(15) \quad \boxed{\begin{aligned} d_0^C &= E - \alpha - \delta, & d_1^C &= \frac{\delta}{R}, & d_2^C &= \delta, \\ d_3^C &= \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\}, & d_4^C &= \alpha, & d_5^C &= \frac{\alpha}{W}. \end{aligned}}$$

Moreover,

$$(16) \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

For closures of original open C triangles one has $\alpha, \delta > 0$.

Remark 2.16 (Affine-chart variables). If $\kappa_j = F_j(O)$ denotes the auxiliary side slack in the CE1 affine chart, take

$$\lambda = R, \quad s = \frac{k}{R}, \quad t = W + \delta, \quad \kappa_0 = \alpha, \quad \kappa_2 = \delta.$$

For CE2, take

$$x = \frac{k}{W}, \quad \nu = R + \alpha, \quad y = \frac{k}{R}, \quad v = W + \delta.$$

If $S = x + y$ and $D = \sqrt{x^2 + xy + y^2}$, then

$$S = \frac{k}{RW}, \quad D = \frac{Ek}{RW},$$

and the old coupling equation is

$$(\nu + v)S - xy = D.$$

Thus the CE1 and CE2 formula lists are two coordinate readings of Proposition 2.15.

Lemma 2.17 (CE2 total slack). *In the CE2 sign range, with $T = \alpha + \delta$,*

$$(17) \quad T < \frac{\eta}{E}.$$

2.2.2. *Boundary contribution and diameter transfer.*

Corollary 2.18 (Common center-boundary formula). *The full center-boundary contribution is*

$$(18) \quad L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}.$$

In CE1,

$$L_{\partial H}(T_C) \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4}.$$

In CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

Lemma 2.19 (Adjacent-edge diameter transfer). *For $0 \leq q \leq 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

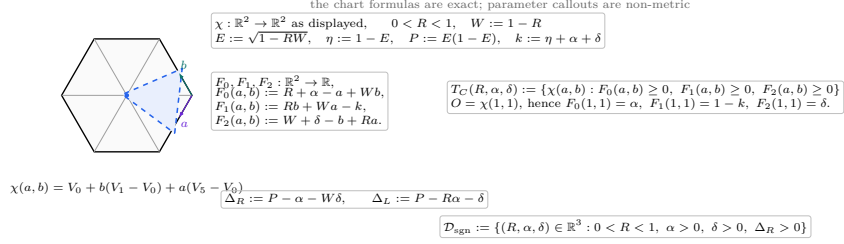
Here the argument-free symbol M_0 continues to denote the radial midpoint; $M_0(q)$ denotes the envelope map. If a unit equilateral triangle reaches parameters q, b on two adjacent boundary edges, then

$$b \leq M_0(q).$$

The map M_0 is strictly decreasing. For $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \quad 1 - M_0(q) > \frac{q}{2}.$$

(e) signed parameters and affine center chart



R is an affine slope parameter; only $\ell_R, \ell_L, d_i^C, c_i^C$ in the trace-and-exit panels are displayed as skeleton lengths.

FIGURE 4. The affine center chart and the exact dependency structure of the signed parameters.

(f) companion-surplus sign on $e_{5,0}$

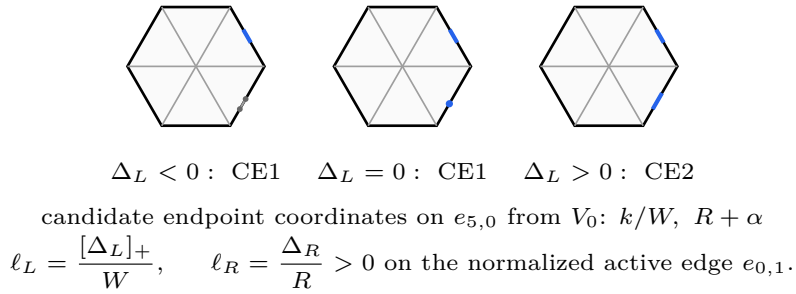
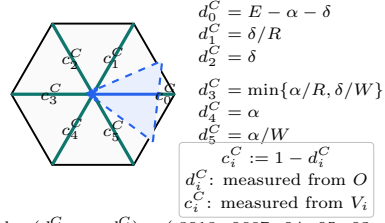
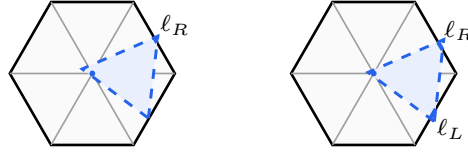


FIGURE 5. The exact sign test for the candidate companion trace, including the CE1 point-contact case.

(h) center exits and complementary radial reach lower bounds

CE2 sample: $(d_0^C, \dots, d_5^C) \approx (.8018, .0667, .04, .05, .03, .075)$.

FIGURE 6. The six exact center exits d_i^C and complementary radial lower bounds $c_i^C = 1 - d_i^C$.

(i) boundary-contribution bounds on the skeleton

$$\begin{aligned} \text{CE1 : } L_{\partial H}(T_C) = \ell_R &\leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4} \\ \text{CE2 : } L_{\partial H}(T_C) = \ell_R + \ell_L &< \frac{1}{2} \end{aligned}$$

FIGURE 7. The exact CE1 and CE2 center-boundary contributions and their general bounds.

3. TRACE-LENGTH BOUNDS

The first mechanism compares the total one-dimensional trace available from the seven triangles with the length of either the perimeter ∂H or the full skeleton S . The local trace estimates, the master perimeter deficit, the conditional skeleton theorem, and every routed application are proved in this section.

3.1. Detailed Trace-Length Estimates. This section develops the boundary and conditional-skeleton trace bounds used by the length mechanism.

For a closed triangle T and a rectifiable target E , write

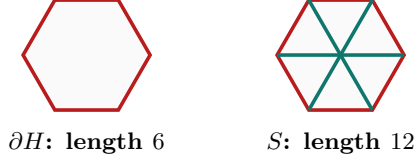
$$L_E(T) = \mathcal{H}^1(T \cap E).$$

The passage from an original open triangle U to its closure $T = \overline{U}$ can only enlarge the trace. Hence a cover of E by the open triangles implies

$$(19) \quad \mathcal{H}^1(E) \leq L_E(T_C) + \sum_{i=0}^5 L_E(T_i).$$

All estimates in this section concern full traces. They therefore remain valid for any assigned measurable subsets of those traces.

The strategy uses only the perimeter ∂H and the full skeleton S , whose measures were recorded in (1). Every skeleton estimate below is conditional on its stated center and V triangle hypotheses; no unconditional noncoverage assertion for S is made.



the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 8. The two one-dimensional targets. The full skeleton is used only under the hypotheses of the corresponding trace budget.

3.1.1. *Conditional skeleton trace caps.* We now develop the skeleton estimate used only when the center is CE1 or CE2. The first lemma is the center contribution.

Lemma 3.1 (Center skeleton cap). *If T_C is a CE1 or CE2 C triangle and contains exactly one radial midpoint, then*

$$L_S(T_C) < \frac{3}{2}.$$

Lemma 3.2 (Positive-support skeleton cap). *Let T_i be the closure of an original open V triangle, so $V_i \in \text{int}(T_i)$. If*

$$\mathcal{H}^1(T_i \cap r_j) > 0 \quad \text{for some } j \in \{i-1, i+1\},$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Lemma 3.3 (No-support skeleton cap). *If a V triangle has $A_i + B_i \leq 1$ and*

$$\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \leq 2.$$

For the remaining cap we use one specialized part of the exact local equilateral enclosure criterion. In local directions u, v meeting at 120° , suppose a closed unit triangle contains

$$0, \quad au, \quad bv, \quad c(u+v),$$

with $0 \leq a \leq b \leq 1$ and $a + b > 1$. The support-line calculation gives the selected supercritical cell

$$(20) \quad \begin{aligned} a^2 + ab + b^2 &\leq 1, & c &\leq \frac{1}{2}, \\ (a^2 - 1)c^2 + (2ab^2 + b)c + (b^4 - b^2) &\leq 0, \end{aligned}$$

where c lies on the connected component containing $c = 0$. To see the selector directly, the quadratic is negative at $c = 0$, positive at $c = 1/2$, and opens downward. For the middle assertion, four times its value at $1/2$ is increasing in a and, at

$a = 1 - b$, equals $b^2(2b - 1)^2$; it is therefore positive when $a + b > 1$. Hence $c \leq 1/2$ selects its smaller-root component. Formula (20) follows by fixing in the support sum the three active contacts $\{B\}$, $\{B, c(u+v)\}$, and $\{A\}$, and squaring the positive unit-side equation.

Lemma 3.4 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Lemma 3.5 (A positive-support adjacent exceptional triangle is forced). *Suppose T_C is CE1 or CE2 and, after symmetry, $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$. If at least two V triangles are supercritical, then for a supercritical index $s \neq 0$ there is some $j \in \{s - 1, s + 1\}$ such that*

$$M_s \in U_j, \quad \mathcal{H}^1(T_j \cap r_s) > 0.$$

3.1.2. *Vd1/Vd2 corner estimates.* We next isolate the much stronger trace loss caused by Vd1 and Vd2 geometry. For the following two results use the temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0).$$

Thus $V_0 = (0, 0)$, $V_5 = (1, 0)$, $V_1 = (0, 1)$, $O = (1, 1)$, and $\|(x, y)\|^2 = x^2 + y^2 - xy$.

Proposition 3.6 (Vd1/Vd2 corner normal form). *Let T be the closure of an original Vd1 or Vd2 triangle at V_0 , and let a, b be its exact reaches toward V_5, V_1 . There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$T = \left\{ (x, y) : \begin{array}{l} x - (t + 1)y \leq a, \\ ty - (t + 1)x \leq tb, \\ tx + y \leq d - a - tb \end{array} \right\}.$$

The raw traces, in coordinates measured from the indicated outer vertex, are

$$\begin{array}{l|l} r_0 & 0 \leq s \leq \frac{d - a - tb}{t + 1} \\ r_1 & \frac{t(1 - b)}{t + 1} \leq s \leq \frac{d - a - tb - 1}{t} \\ r_5 & \frac{1 - a}{t + 1} \leq s \leq d - a - tb - t. \end{array}$$

Intersecting with $[0, 1]$ gives the actual traces. Moreover,

$$(21) \quad a + b < \frac{1}{2}.$$

Lemma 3.7 (Vd2 adjacent-midpoint cap). *If an open role U_i has Vd2 closure $T_i = \overline{U_i}$ and T_i contains M_{i-1} or M_{i+1} , then its exact boundary reaches satisfy*

$$a + b < \frac{1}{3}.$$

3.1.3. *Boundary trace caps.* The signed center normal form gives the CE1 and CE2 center contributions from one formula; no separate maximal-interval calculation is needed here.

Theorem 3.8 (Boundary trace table). *For closures of original open triangles, the following bounds hold:*

<i>triangle</i>	<i>hypothesis</i>	$L_{\partial H}$
T_C	CE0	0
T_C	CE1	$\leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
T_C	CE2	$< \frac{1}{2}$
T_i	Vd0, $A_i + B_i \leq 1$	≤ 1
T_i	Vd0, $A_i + B_i > 1$	$\leq \frac{2}{\sqrt{3}}$
T_i	Vd1/Vd2	$< \frac{1}{2}$
T_i	T3-like	< 1 .

3.1.4. *Final length contradictions.*

Proposition 3.9 (Vd2 adjacent-midpoint branch). *Assume a CE2, $N_+ = 1$ configuration has exactly one Vd1/Vd2 triangle, that triangle is T_i of type Vd2 and contains M_{i-1} or M_{i+1} , and all other V triangles are Vd0. Then the boundary cannot be covered.*

Proposition 3.10 (Branches closed by Method 1). *Direct length-sum obstructions close the following routing entries and hybrid subcase:*

- (1) $N_{\text{gap}} = 0$, $N_+ = 0$, for any center and V triangle types;
- (2) $N_{\text{gap}} = 0$, $N_+ = 1$, with a Vd1/Vd2 triangle;
- (3) CE1/CE2 with $N_+ + N_{\text{sp}} \geq 3$;
- (4) CE1/CE2, $N_+ = 0$, with a Vd1/Vd2 triangle;
- (5) CE1, $N_+ = 1$, with exactly one Vd1/Vd2 triangle and no T3-like triangle;
- (6) the CE2 hybrid subcase in which the Vd2 triangle T_i contains M_{i-1} or M_{i+1} .

3.2. Common perimeter and skeleton budgets. The signed center formula also removes repeated length arithmetic. We first state a generic perimeter deficit, then give the direct CE1/CE2 skeleton count used by the routing table.

Lemma 3.11 (Master perimeter deficit). *Put*

$$\theta = \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has a center contribution L_C , n supercritical Vd0 V triangles, $m \geq 1$ distinguished nonsupercritical V triangles with strict boundary caps $q_1, \dots, q_m < 1$, and $6 - n - m$ remaining V triangles whose boundary contributions are at most 1. If

$$(22) \quad L_C + n\theta \leq \sum_{j=1}^m (1 - q_j),$$

then the seven triangles do not cover ∂H .

Lemma 3.12 (Small center slacks in the one-Vd1/Vd2 branch). *Assume there is exactly one supercritical Vd0 V triangle, exactly one Vd1/Vd2 V triangle, and four*

nonsupercritical Vd0 V triangles. Then every candidate surviving the perimeter budget is CE2 and satisfies

$$(23) \quad \boxed{\alpha + \delta < \frac{1}{24}, \quad \alpha + \delta < \frac{\min\{R, W\}}{6}.}$$

Lemma 3.13 (CE2 initial endpoints dominate the total slack). *Assume $\Delta_R > 0$ and $\Delta_L > 0$. Put*

$$T = \alpha + \delta, \quad x = \frac{\eta + T}{W}, \quad y = \frac{\eta + T}{R}.$$

Then

$$(24) \quad \boxed{T < \frac{1}{2} \min\{x, y\}.}$$

Theorem 3.14 (Direct CE1/CE2 skeleton count). *If a CE1/CE2 branch satisfies $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$ and*

$$(25) \quad N_+ + N_{\text{sp}} \geq 3,$$

then the seven triangles do not cover the full skeleton S .

Corollary 3.15 (Length branches closed by the common budgets). *The following branches are impossible.*

- (1) CE0, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;
- (2) $N_+ = 0$ and at least one Vd1/Vd2 V triangle, for either CE1 or CE2;
- (3) CE1, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;
- (4) CE2, $N_+ = 1$, and at least two Vd1/Vd2 V triangles;
- (5) CE2, $N_+ = 1$, with a Vd2 triangle T_i containing M_{i-1} or M_{i+1} ;
- (6) CE1 or CE2 with $N_+ + N_{\text{sp}} \geq 3$;
- (7) CE1 or CE2 with $N_+ \geq 2$.

4. AREA-LOSS ESTIMATE

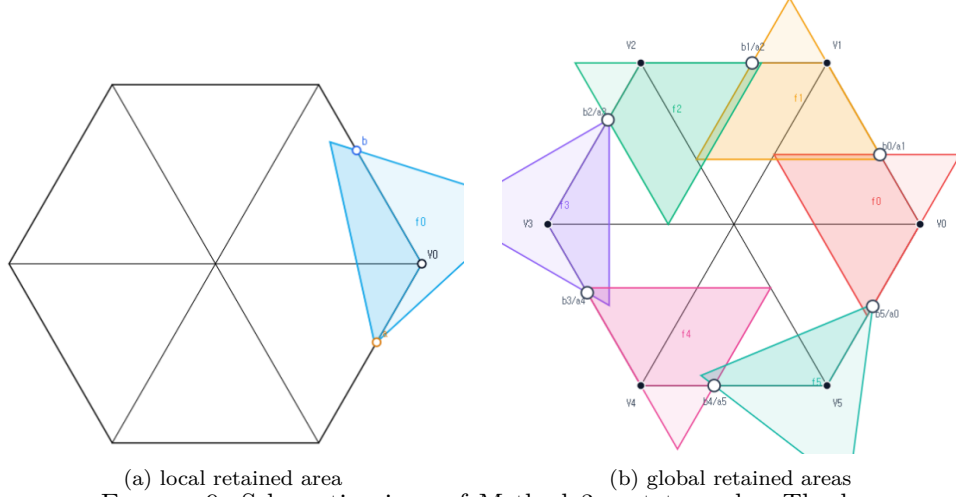
The second strategy measures the normalized area forced outside H by a V triangle with prescribed boundary reaches. The local square-loss and T3-like estimates are proved first, then summed around the six-cycle to close the two center-independent zero-gap area branches.

4.1. Detailed Area-Loss Estimates. This section proves the local inequalities and cyclic estimates used in the area-loss argument.

Let

$$A_\Delta = \frac{\sqrt{3}}{4}$$

be the area of a unit equilateral triangle. Throughout this section areas are divided by A_Δ . Thus a unit triangle has normalized area one and H has normalized area six.



(a) local retained area (b) global retained areas
 FIGURE 9. Schematic views of Method 3, not to scale. The local panel shows one V triangle with boundary reaches a, b , and the global panel shows the six cyclic triangles. Only the darker portions represent area retained inside H ; the proof uses the inequalities below, not the picture.

4.1.1. *The local square-loss inequalities.* Fix a vertex V_i . Let P_a and P_b be the points at distances a and b from V_i on its incoming and forward boundary edges. For a closed unit equilateral triangle T containing V_i, P_a, P_b , put

$$\mathcal{L}(T) = \frac{\text{area}(T \setminus H)}{A_\Delta}.$$

Equivalently, if

$$f(a, b) = \max_T \frac{\text{area}(T \cap H)}{A_\Delta}, \quad \mathcal{L}(a, b) = 1 - f(a, b),$$

then $\mathcal{L}(a, b)$ is the least possible normalized loss. Reflection in the radial axis interchanges a and b , and increasing either reach lower bound can only decrease f .

Lemma 4.1 (Local wedge reduction). *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \quad W = \{(x, y) : x \geq 0, y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0 , then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x, y) -coordinate area.

Theorem 4.2 (Unconditional local square loss). *For every feasible pair $0 \leq a, b \leq 1$ and every realizing closed unit equilateral triangle T ,*

$$(26) \quad \mathcal{L}(T) \geq \min(a, b)^2.$$

If $a + b > 1$, then

$$(27) \quad \mathcal{L}(T) \geq \max(a, b)^2.$$

Consequently the same inequalities hold for $\mathcal{L}(a, b) = 1 - f(a, b)$.

4.1.2. *The direct T3-like loss.* The next theorem uses the classification data (o, n) from Section 2.1: o counts triangle vertices outside H , and $n = n(T_i)$ is

$$|\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|.$$

Thus a T3-like triangle has $(o, n) = (2, 1)$.

Theorem 4.3 (Direct T3-like loss). *Let T be a closed unit V_i -triangle of T3-like type, and let a, b be lower bounds for its adjacent boundary reaches. Then*

$$(28) \quad a + b \leq 1.$$

Moreover, for $0 \leq m \leq 1/2$,

$$(29) \quad a, b \geq m \implies \mathcal{L}(T) \geq 2m - 4m^2.$$

4.1.3. *The cyclic area certificates.* The selected handoffs of Proposition 2.10 have the form

$$(a_i, b_i) = (1 - \xi_{i-1}, \xi_i), \quad 0 < \xi_i < 1,$$

with cyclic indices. V triangle i is selected supercritical exactly when $\xi_i > \xi_{i-1}$.

Lemma 4.4 (Cyclic area-loss certificate). *Let the six actual V triangles realize selected handoffs*

$$(a_i, b_i) = (1 - \xi_{i-1}, \xi_i), \quad 0 < \xi_i < 1.$$

Their total normalized loss is strictly greater than one if either

- (1) *at least two selected V triangles are supercritical; or*
- (2) *exactly one selected V triangle is supercritical, every triangle is Vd0 or T3-like, and at least one triangle is T3-like.*

Proposition 4.5 (Branches closed by area). *The following branches are impossible:*

- (1) $N_{\text{gap}} = 0$, $N_+ \geq 2$, *with arbitrary center and V triangle types;*
- (2) $N_{\text{gap}} = 0$, $N_+ = 1$, *no triangle is Vd1 or Vd2, and at least one triangle is T3-like.*

5. FINITE-ENCLOSURE OBSTRUCTIONS

The finite-enclosure strategy forces explicit points or compact sets into the open C triangle. Global rows end with an enclosing-side inequality $\Lambda \geq 1$; local rows end when a forced point lies beyond a signed C-triangle exit. The simplified proof uses three recurring forms:

common radial disk + actual gap,

two transverse radial points in CE2,

and

$$K_{\text{tr}} = \{O, M_0, X(\ell), X(r), P_2, P_3, P_4\}$$

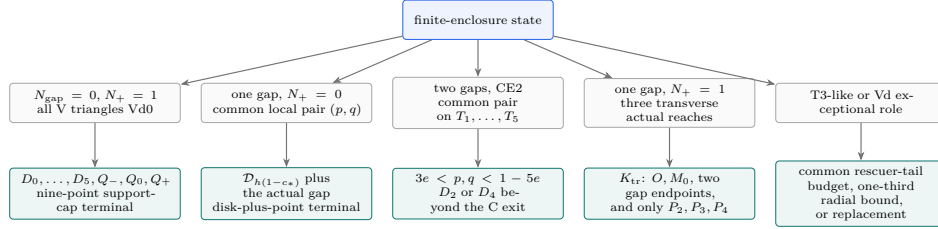
for the one-gap, one-supercritical all-Vd0 row.

normalized subcase	center-forced object	terminal
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5.1. Finite-enclosure case summary.

Table 2: Finite-enclosure subcases after the simplifications.

normalized subcase	center-forced object	terminal
$N_{\text{gap}} = 0, N_+ = 1$, all Vd0	the radial disk and Q_-, Q_0, Q_+	the exact nine-point theorem
$N_{\text{gap}} = 1, N_+ = 0$, all Vd0 or at most two T3-like roles	$\mathcal{D}_{h(1-c*)}$ and $J(p, q)$	the complementary-gap theorem
$N_{\text{gap}} = 2, N_+ \in \{0, 1\}$, all Vd0/T3-like	D_2, D_4	the shortened $3e - (1 - 5e)$ CE2 theorem
$N_{\text{gap}} = 1, N_+ = 1$, all Vd0	$K_{\text{tr}} = \{O, M_0, X(\ell), X(r), P_2, P_3, P_4\}$	endpoint upper squeeze versus the three-transverse return
$N_{\text{gap}} = 1, N_+ = 1$, exactly one T3-like	the O-side endpoint P_T	the common rescuer-tail budget
one Vd role adjacent to supercritical T_0	a point on r_2	the half-edge one-third radial envelope
one Vd role nonadjacent to supercritical T_0	D_τ	direct separation from both Vd and C traces
Vd1 based at T_0	the O-side Vd1 endpoint	the common rescuer-tail budget
Vd1-supercritical pair away from T_0	the corrected all-Vd0 replacement output	rerouting by the recomputed gap rank



The simplified routes retain only the support data used by their terminal.

FIGURE 10. Roadmap of the simplified finite-enclosure chapter. The one-gap all-Vd0 witness uses only three transverse radial endpoints; the two-gap terminal uses the interval $3e < p, q < 1 - 5e$; and the T3-like and Vd1 paths share one rescuer-tail lemma.

5.2. Trace-exact AB envelopes at an actual gap. If the actual V-gap on $e_{i,i+1}$ is

$$J_i = X_i([\ell, r]), \quad X_i(t) = V_i + t(V_{i+1} - V_i),$$

then

$$(6.24a) \quad \boxed{\ell = B_i, \quad r = 1 - A_{i+1}.$$

Consequently

$$(6.24b) \quad U_i \cap e_{i,i+1} = X_i([0, \ell]), \quad U_{i+1} \cap e_{i,i+1} = X_i((r, 1]),$$

and the entire closed interval J_i , including a singleton, is center-forced.

For a closed unit V triangle S , define the source-conditioned unions

$$(6.24c) \quad \mathcal{E}_i^{\rightarrow}(a \mid b) = \bigcup \{S \cap \mathcal{C}_i : A_i(S) \geq a, B_i(S) = b\},$$

$$(6.24d) \quad \mathcal{E}_i^{\leftarrow}(a | b) = \bigcup \{S \cap \mathcal{C}_i : A_i(S) = a, B_i(S) \geq b\}.$$

Their relevant edge sections stop and start exactly at the actual gap endpoints, and

$$(6.24e) \quad \mathcal{E}_i^{\rightarrow}(a | b), \quad \mathcal{E}_i^{\leftarrow}(a | b) \subseteq \mathcal{K}_i(a, b).$$

Thus the ordinary capacities remain valid while endpoint ownership is exact.

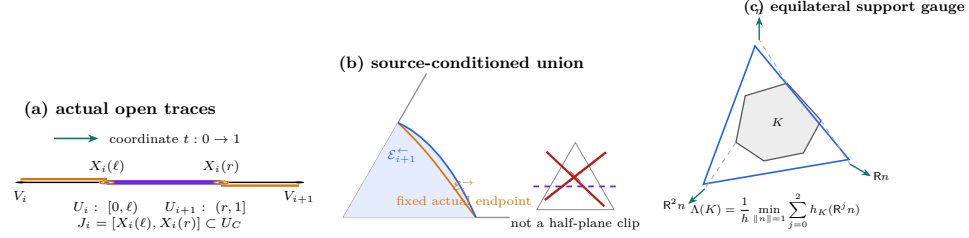


FIGURE 11. Three conventions used throughout the finite-enclosure argument. (a) A hollow outer marker records nonmembership in an open V trace, while the filled center marker records selection of the same endpoint as a center-forced witness. (b) A trace-exact envelope is obtained by restricting the source family to the prescribed actual maximal reach and then taking its union; it is not an affine half-plane clipping. (c) Equilateral enclosure is measured by three support directions separated by 120° . Incidences and interval orders are exact; lengths are schematic.

5.3. Direct local enclosure calculus. Let R denote counterclockwise rotation through $2\pi/3$, and put $h = \sqrt{3}/2$. For a nonempty compact set K , let

$$h_K(n) = \max_{x \in K} \langle x, n \rangle.$$

Proposition 5.1 (Equilateral enclosure gauge). *The least side length of a closed equilateral triangle containing K is*

$$(6.1) \quad \Lambda(K) = \frac{1}{h} \min_{\|n\|=1} \sum_{j=0}^2 h_K(R^j n).$$

If K is a polygon, a minimizing normal can be chosen perpendicular to a hull edge.

Take unit vectors u, v meeting at 120° , so that $u+v$ is the inward radial direction, and put

$$K(a, b, c) = \text{conv}\{0, au, bv, c(u+v)\}.$$

The demand triple (a, b, c) is *admissible* when $K(a, b, c)$ lies in a closed unit equilateral triangle.

Proposition 5.2 (Exact local admissible set). *Put*

$$s = a + b, \quad d = a^2 + ab + b^2, \quad m = \min\{a, b\}, \quad M = \max\{a, b\}, \\ q = s^4 - s^2 + ab,$$

and

$$\begin{aligned} F_L &= c^4 - c^2 + mc - m^2, \\ F_T &= (s^2 - 1)c^2 + Mc - M^2, \\ F_S &= (m^2 - 1)c^2 + (2mM^2 + M)c + M^4 - M^2. \end{aligned}$$

Then the admissible set is the union of

$$\begin{aligned} \mathcal{A}_L &= \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\}, \\ \mathcal{A}_T &= \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\}, \\ \mathcal{A}_S &= \{d \leq 1, s > 1, F_S \leq 0, c \leq 1/2\}. \end{aligned}$$

Consequently the maximal admissible radial demand is

$$(6.2) \quad c_{\max}(a, b) = \begin{cases} 1, & a = b = 0, \\ C_L(m), & s \leq 1, q \leq 0, \quad (a, b) \neq (0, 0), \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & s \leq 1, q > 0, \\ \frac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}, & s > 1, \end{cases}$$

where $C_L(m)$ is the selected root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

The admissible set is coordinatewise down-closed.

The zero-radial diameter envelope is

$$(6.3) \quad M_0(x) = \frac{-x + \sqrt{4 - 3x^2}}{2}.$$

For

$$0 < e < 1 - \frac{\sqrt{3}}{2},$$

define the selected low root

$$(6.4) \quad \epsilon(e) = \frac{1 - e}{2} \left(1 - \sqrt{4(1 - e)^2 - 3} \right).$$

It satisfies

$$(6.5) \quad e < \epsilon(e), \quad \epsilon(e) < \frac{2e}{1 - 2e},$$

and, for $e \leq 1/8$,

$$(6.6) \quad \epsilon(e) \leq 2e + 5e^2.$$

These follow by substituting the proposed upper bounds in the selected quadratic and checking its sign.

Lemma 5.3 (Direct high-radial threshold). *Let a nonsupercritical V triangle have incoming boundary reach A and outgoing boundary reach B . If*

$$C \geq 1 - e, \quad A > \epsilon(e),$$

then

$$(6.7) \quad \boxed{B \leq \epsilon(e)}.$$

For $0 \leq c \leq 1/2$, the strict-supercritical outgoing supremum is

$$(6.8) \quad M_c^{\sup} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

It is not attained. Thus an actual strict-supercritical V triangle with own-radial reach at least c has outgoing boundary reach strictly below $M_c^{\sup}$.

Lemma 5.4 (Selected-arc chord bounds). *Let $0 < e < 1/2$. On the selected Q_+ component for radial demand $1 - e$, let p be the incoming boundary reach and let q be the following incoming reach forced across a center-free edge. Then*

$$q - p = 1 - \sqrt{1 - x + x^2},$$

where

$$(6.9) \quad q = e + (1 - e)x, \quad p = e + (1 - e)x - 1 + \sqrt{1 - x + x^2}.$$

The selected arc is increasing and concave. In particular, on every interval used below,

$$(6.10) \quad q \geq p + \lambda(p - e)$$

for the chord slope λ joining the appropriate selected endpoints. The CE1 applications may take

$$(6.11) \quad \lambda > 1 - 4e \quad (e = \alpha), \quad \lambda > 1 - 5e \quad (e = \alpha/R).$$

5.4. Exact neighboring-ray capacity. The T3-like and Vd placements require the exact capacity of a V triangle on a neighboring radial arm. Normalize at V_0 , let $A_a \in e_{5,0}$ and $B_b \in e_{0,1}$ have reaches a, b , and let

$$X_c = (1 - c)V_1 \in r_1.$$

Define $C_+(a, b)$ as the maximal c for which V_0, A_a, B_b, X_c lie in a closed unit equilateral triangle. Put $C_-(a, b) = C_+(b, a)$.

For $0 \leq a \leq 1/2$, let $p(a)$ be the unique root in $[a, 1]$ of

$$(6.24a) \quad p^3 - (a + 2)p^2 + 2(a + 1)p - 1 = 0,$$

and define

$$\sigma(a) = 1 - p(a), \quad \tau(a) = 1 - a - (p(a) - a)(1 - p(a)).$$

Proposition 5.5 (Neighboring-ray formula).

$$(6.24b) \quad C_+(a, b) = \begin{cases} \text{undefined}, & a + b > 1, \\ 1 - b, & a + b \leq 1, \ a > 1/2, \\ 1 - b, & 0 \leq a \leq 1/2, \ 0 \leq b \leq \sigma(a), \\ p(a), & 0 \leq a \leq 1/2, \ \sigma(a) \leq b \leq \tau(a), \\ a + \frac{1}{2} - \sqrt{(a + b)^2 - \frac{3}{4}}, & 0 \leq a \leq 1/2, \ \tau(a) < b \leq 1 - a. \end{cases}$$

At $b = \tau(a)$ the plateau value $p(a)$ is selected.

A consequence used repeatedly below is

$$(6.24e) \quad \boxed{C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\}} \quad (p + q \leq 1).$$

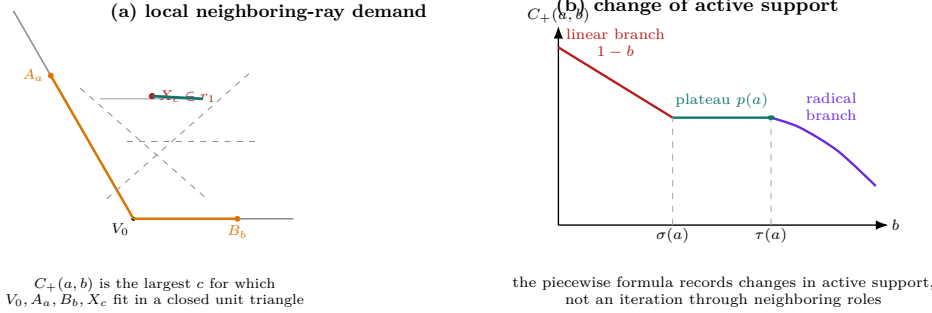


FIGURE 12. Geometry behind the neighboring-ray capacity $C_+(a, b)$. The local problem asks how far a triangle anchored at V_0 and the two boundary points A_a, B_b can reach on the neighboring arm r_1 . As b varies with a fixed, the active support pattern changes from the linear branch to a plateau and then to the radical branch, with transition points $\sigma(a)$ and $\tau(a)$. The graph is schematic; the transition order, endpoint selection, and branch labels are exact.

Indeed this is immediate on the linear branch; on the plateau branch the cubic gives $p(a) \leq 1 - a$, and the radical branch is no larger than its selected plateau at the transition. **Simplified finite-enclosure interfaces.**

Proposition 5.6 (Disk–finite-set calipers). *Let $\mathcal{D}_\eta$ be centered at O , let $P = \{p_1, \dots, p_m\}$, and let J denote rotation by $\pi/2$. Unless the disk is the active support in all three directions at a minimizing orientation, a minimizing normal belongs to the finite list*

$$\pm \frac{J(p_k - p_i)}{\|p_k - p_i\|} \quad (p_i \neq p_k)$$

or

$$\frac{\eta p_i \pm \sqrt{\|p_i\|^2 - \eta^2} J p_i}{\|p_i\|^2} \quad (\|p_i\| \geq \eta).$$

Thus one side of a least enclosing equilateral triangle either contains two points, or is tangent to the disk and contains one point.

Lemma 5.7 (Half-edge one-third radial envelope). *If*

$$M \geq \frac{1}{2}, \quad 0 < m \leq M, \quad M + m < 1,$$

then

$$(6.34a) \quad \boxed{c_{\max}(M, m) < 1 - \frac{m}{3}}.$$

Lemma 5.8 (Rescuer endpoint and boundary-tail budget). *Suppose a special role at V_0 has boundary endpoint a on $e_{5,0}$ and supported interval $[c, u] \subset r_1$, measured from V_1 toward O . Put*

$$\varepsilon = 1 - u, \quad M = M_c^{\sup}.$$

Assume

$$\varepsilon V_1 \in U_C, \quad a + \varepsilon \leq 1,$$

$$(6.34b) \quad a \leq 1 - M, \quad \frac{a}{a + \varepsilon} \leq 1 - M.$$

If T_1 is uniquely supercritical with own-radial reach at least c and T_2, T_3, T_4, T_5 are nonsupercritical on the center-free four-edge path, then the skeleton is not covered.

5.5. Type-aware radial witnesses. Suppose $T_i = \overline{U_i}$ contains selected boundary anchors with $a_i \leq A_i$ and $b_i \leq B_i$. Put

$$\Gamma_i = \max\{c_{\max}(a_i, b_i), C_+(a_{i-1}, b_{i-1}), C_-(a_{i+1}, b_{i+1})\},$$

omitting neighboring terms forbidden by the actual V types, and define

$$(6.25) \quad D_i = (1 - \Gamma_i)V_i.$$

Lemma 5.9 (Type-aware radial forcing). *Every D_i is missed by all six open V roles and therefore lies in U_C .*

Lemma 5.10 (Common-pair domination). *Let $p, q \geq 0$, $p + q \leq 1$, and put*

$$c_* = c_{\max}(p, q), \quad m = \min\{p, q\}.$$

Then

$$(6.26) \quad c_* \geq 1 - m, \quad C_+(p, q), C_-(p, q) \leq 1 - m \leq c_*.$$

The same conclusion holds for every actual pair dominating (p, q) .

5.6. A disk and one boundary point.

Lemma 5.11 (Disk-plus-point formula). *Let $\mathcal{D}_\eta$ be centered at O , and let $\|G\| = \rho$. Then*

$$(6.27) \quad \Lambda(\mathcal{D}_\eta \cup \{G\}) = \begin{cases} 3\eta/h, & \rho \leq 2\eta, \\ \frac{3\eta + \sqrt{3(\rho^2 - \eta^2)}}{2h}, & \rho > 2\eta. \end{cases}$$

For $p + q \leq 1$, put

$$J(p, q) = [X(q), X(1 - p)] \subset e_{0,1}, \quad c_* = c_{\max}(p, q), \quad d = 1 - c_*.$$

Theorem 5.12 (Complementary-gap enclosure).

$$(6.28) \quad \boxed{\Lambda(\mathcal{D}_{hd} \cup J(p, q)) \geq 1.}$$

5.7. The shortened CE2 two-gap theorem.

Theorem 5.13 (CE2 short-ray obstruction). *In the strict signed CE2 domain put*

$$p = W - \alpha, \quad q = R - \delta, \quad e = \min\{\alpha, \delta\}.$$

Then

$$(6.30) \quad \boxed{c_{\max}(p, q) < 1 - e.}$$

5.8. Nonzero-gap cases with $N_+ = 0$.

Proposition 5.14 (All-Vd0 and T3-like gap closures). *Every nonzero-gap state with $N_+ = 0$, no Vd1/Vd2 role, and at most two T3-like roles is impossible.*

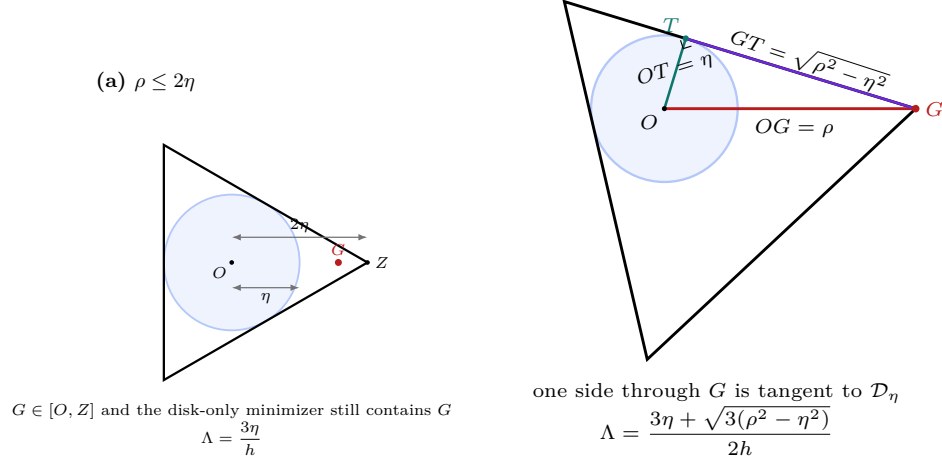
(b) $\rho > 2\eta$ 

FIGURE 13. The two minimizing configurations in the disk-plus-point formula. For $\rho \leq 2\eta$, the centered equilateral triangle tangent to $\mathcal{D}_\eta$ already contains G . For $\rho > 2\eta$, the point G becomes a vertex of the minimizing triangle and one incident side is tangent to the disk. The tangent length $GT = \sqrt{\rho^2 - \eta^2}$ is the geometric source of the radical term.

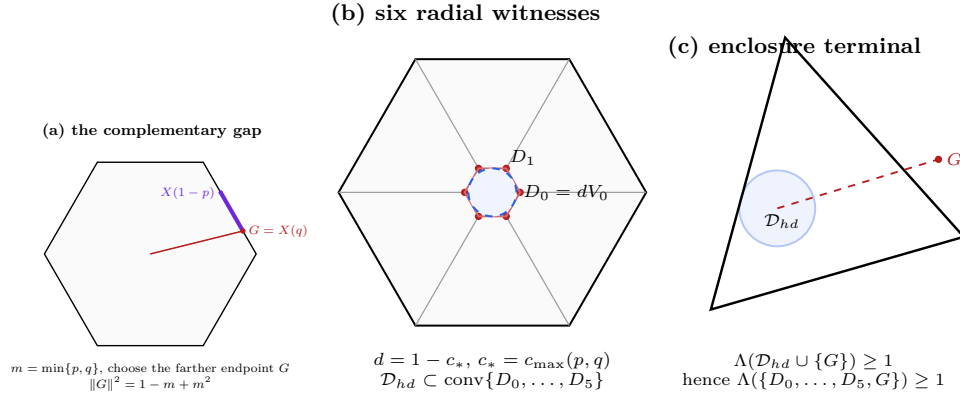


FIGURE 14. The complementary-gap enclosure mechanism. The farther endpoint G of $J(p, q)$ is center-forced. Common-pair domination produces the six center-forced points $D_i = (1 - c_*)V_i$, whose convex hull contains the disk $\mathcal{D}_{h(1-c_*)}$. The disk-plus-point formula then forces enclosing side length at least one. The disk is not an additional witness; it is contained in the convex hull of the six finite radial witnesses.

(a) two CE2 gaps

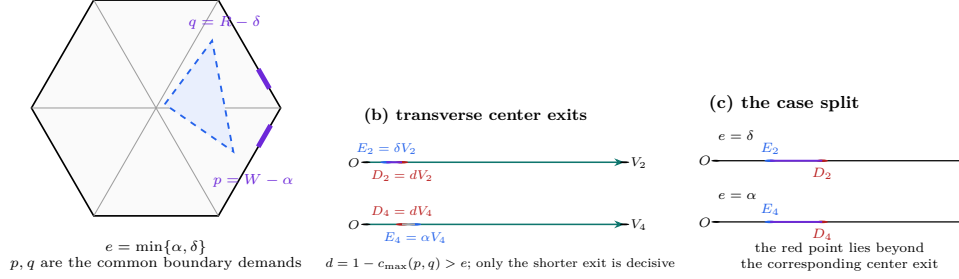


FIGURE 15. The CE2 short-ray terminal. The two boundary gaps determine the common pair (p, q) and the shorter center residual $e = \min\{\alpha, \delta\}$. The theorem proves $d = 1 - c_{\max}(p, q) > e$. Consequently the radial witness on the shorter of r_2 and r_4 lies strictly beyond the corresponding center exit and is uncovered. The displayed numerical ordering is illustrative; the labelled case split is exact.

5.9. The CE1 one-gap reverse-path certificate. The CE1 part of the seven-point enclosure theorem requires one longer direct calculation. We state the exact interface separately: only five boundary handoffs and the three transverse radial reaches at T_4, T_3, T_2 are used.

Proposition 5.15 (CE1 three-transverse return certificate). *Use the signed CE1 center variables*

$$0 < R < 1, \quad w = 1 - R, \quad E = \sqrt{1 - Rw},$$

$$\eta = 1 - E, \quad P = E(1 - E),$$

and write

$$A = \alpha, \quad D = \delta.$$

Assume

$$(6.12) \quad A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P.$$

Put

$$(6.13) \quad X = R - D, \quad m = \frac{A}{R}, \quad h_0 = \frac{\eta + A + D}{2R}.$$

Let T_0 be the unique supercritical role and let $T_1, \dots, T_5$ be nonsupercritical Vd0 roles. Assume

$$A_1 \geq X,$$

$$B_i + A_{i+1} \geq 1 \quad (1 \leq i \leq 4), \quad B_5 + A_0 \geq 1,$$

and

$$(6.14) \quad C_4 \geq 1 - A, \quad C_3 \geq 1 - m, \quad C_2 \geq 1 - D.$$

Then

$$(6.15) \quad \boxed{A_0 > 1 - h_0.}$$

CE1 reverse path: from a deficient terminal to the final contradiction

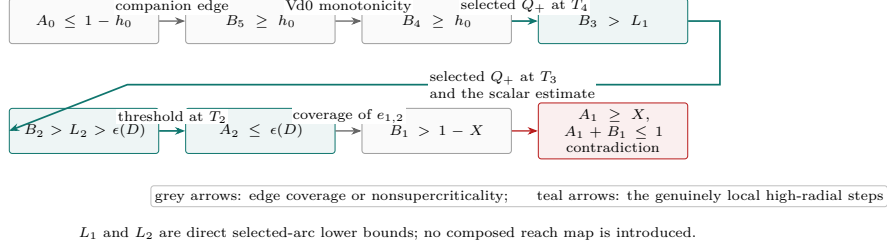


FIGURE 16. Logical structure of the CE1 direct reverse-path certificate. The proof starts from the negation of the desired terminal bound at T_0 , moves through the actual boundary reaches, uses two selected-arc estimates and one high-radial threshold, and returns to T_1 with incompatible incoming and outgoing demands.

5.10. **The one-gap all-Vd0 case.** Let the actual gap be $[X(\ell), X(r)] \subset e_{0,1}$, and let C_i be the actual own-radial reach of T_i . Put

$$P_i = (1 - C_i)V_i.$$

Lemma 5.16 (The transverse witness is center-forced). *Every P_i is missed by all six open V roles. In particular,*

$$(6.33) \quad K_{\text{tr}} = \{O, M_0, X(\ell), X(r), P_2, P_3, P_4\} \subset U_C.$$

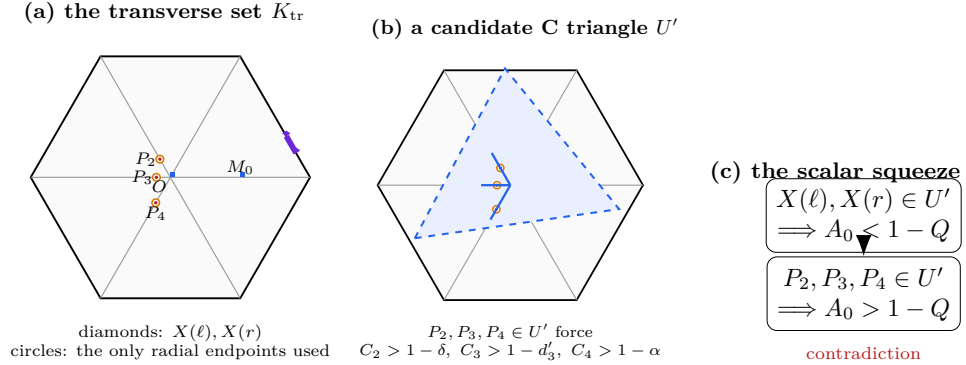


FIGURE 17. The simplified one-gap all-Vd0 enclosure certificate. Only the actual gap endpoints and the three transverse radial endpoints are needed. The gap gives the diameter upper squeeze at the supercritical role; the three radial endpoints supply the CE1/CE2 transverse return. The former points P_0, P_1, P_5 are unnecessary. Lengths are schematic.

Lemma 5.17 (Endpoint upper squeeze). *If an open unit triangle U' contains K_{tr} , give it signed center variables and put*

$$X = R - \delta, \quad Q = \frac{\eta + \alpha + \delta}{2R}.$$

Then

$$(6.34) \quad \boxed{A_0 < 1 - Q.}$$

Proposition 5.18 (CE2 transverse threshold terminal). *No CE2 open unit triangle contains K_{tr} .*

Proposition 5.19 (CE1 three-transverse return). *No CE1 open unit triangle contains K_{tr} .*

Proposition 5.20 (Transverse seven-point enclosure).

$$(6.35) \quad \boxed{\Lambda(K_{\text{tr}}) \geq 1.}$$

Consequently the former ten-point superset K_{410} also has enclosure number at least one.

For two gaps, Theorem 5.13 applies independently of the criticality of T_0 .

5.11. Exactly one T3-like role.

Proposition 5.21 (Direct T3-like rescuer witness). *Assume $N_+ = 1$, exactly one V triangle is T3-like, no V triangle is Vd1/Vd2, and at least one V-gap is present. Then the skeleton is not covered.*

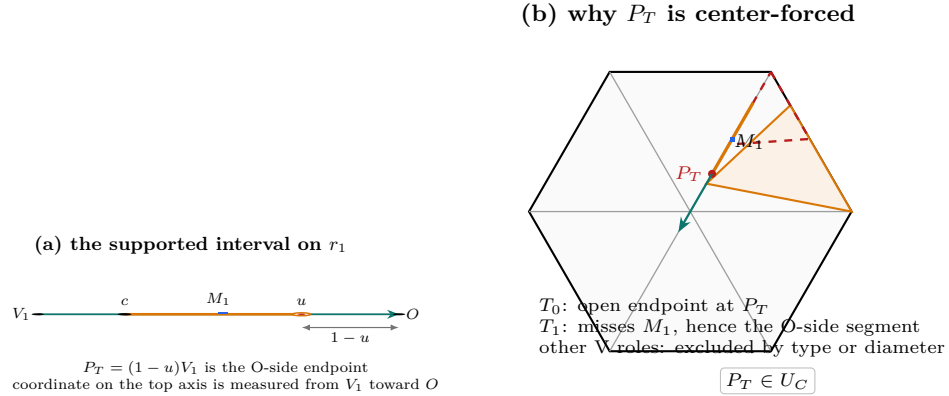


FIGURE 18. Construction of the T3-like rescuer endpoint. The interval $[c, u]$ is parametrized from V_1 toward O , so its O-side endpoint is $P_T = (1 - u)V_1$. The T3-like open role omits its endpoint, the adjacent supercritical role contains V_1 but misses M_1 and therefore misses the O-side segment by convexity, and all remaining V roles are excluded. Thus P_T is forced into the C triangle.

5.12. Exactly one Vd1/Vd2 role in CE2.

Proposition 5.22 (Direct one-Vd placement assembly). *Assume CE2, $N_+ = 1$, exactly one Vd1/Vd2 role, and at least one V-gap. Then the skeleton is not covered.*

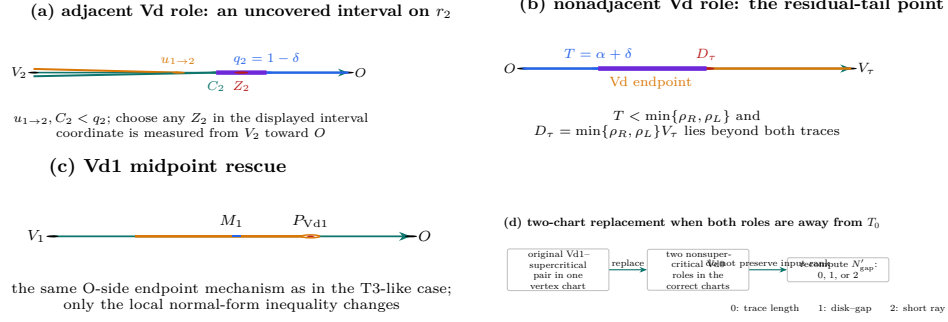


FIGURE 19. The four geometric terminals in the one-Vd placement assembly. In the adjacent placement the radial coordinate is measured from V_2 toward O ; the two local V endpoints lie before the center interval entry $q_2 = 1 - \delta$, leaving a literal uncovered interval on r_2 . The nonadjacent placement produces the point $D_\tau = \min\{\rho_R, \rho_L\}V_\tau$ beyond both the Vd and center traces; the Vd1 midpoint rescue uses the same O-side endpoint construction as the T3-like case; and the remaining placement is reduced by a two-chart replacement followed by recomputation of the output gap rank. Lengths are schematic; all labelled order relations are the proof relations.

5.13. Detailed direct certificates for the nonzero-gap terminals. This subsection records the calculations suppressed by the case proofs. The new CE2 short-ray calculation replaces the former root-separation argument; the remaining residual and local normal-form calculations are unchanged.

5.13.1. The disk-gap support calculation. Let $c = c_{\max}(p, q)$, $m = \min\{p, q\}$, and $d = 1 - c$. The regular radial hexagon has inradius hd , while the farther endpoint G of the complementary gap has

$$\|G\|^2 = 1 - m + m^2.$$

The disk-plus-point formula shows that

$$\Lambda(\mathcal{D}_{hd} \cup \{G\}) \geq 1$$

is equivalent in the nontrivial branch to

$$(6.51) \quad 3d(1 - d) \geq m(1 - m).$$

For $c > h$, the exact local support inequalities imply

$$(6.52) \quad m \leq c \left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}} \right).$$

Substitution gives

$$\begin{aligned} & 3c(1 - c) - c \left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}} \right) \left[1 - c \left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}} \right) \right] \\ &= \frac{c(1 - c)}{2} \left(5 - 2c - 2c^2 + 2\sqrt{c^2 - \frac{3}{4}} \right) \geq 0. \end{aligned}$$

This proves the complementary-gap inequality.

5.13.2. *The CE2 one-gap threshold inequalities.* Put

$$X = R - \delta, \quad Q = \frac{\eta + \alpha + \delta}{2R}.$$

The right-trace inequality gives

$$(6.53) \quad WX = RW - W\delta = \eta + P - W\delta > \eta + \alpha.$$

The selected low root $\epsilon(\alpha)$ is the smaller root of

$$z^2 - (1 - \alpha)z + (1 - \alpha)^2 - (1 - \alpha)^4 = 0.$$

Testing this quadratic at the lower bound in (6.53) is equivalent to the positivity of

$$\pi(q) = (\eta + q)(1 - 2q) - 2qW.$$

The function is concave and

$$\pi(0) = \eta > 0, \quad \pi(P) = \eta(\eta + 2ER^2) > 0.$$

Thus

$$(6.54) \quad \boxed{\epsilon(\alpha) < X.}$$

For the second inequality put $T = \alpha + \delta$. Multiplying the two CE2 inequalities by W, R and adding gives

$$(6.55) \quad E^2T < P = E\eta, \quad T < \frac{\eta}{E}.$$

If $d = \min\{\alpha, \delta\}$, then

$$\min\{\epsilon(\alpha), \epsilon(\delta)\} < \frac{2d}{1 - 2d} \leq \frac{T}{1 - T}.$$

The comparison $T/(1 - T) < Q$ is equivalent to

$$\chi(T) = (\eta + T)(1 - T) - 2RT > 0.$$

The function χ is concave,

$$\chi(0) = \eta > 0, \quad \chi\left(\frac{\eta}{E}\right) = \frac{\eta}{E^2}(E - R)^2 > 0.$$

Hence

$$(6.56) \quad \boxed{\min\{\epsilon(\alpha), \epsilon(\delta)\} < Q.}$$

5.13.3. *The shortened CE2 two-gap calculation.* Let

$$p = W - \alpha, \quad q = R - \delta, \quad e = \min\{\alpha, \delta\}, \quad M = \max\{R, W\}.$$

The strict CE2 inequalities give

$$P > (1 + M)e.$$

Since $E^2 = 1 - M + M^2$,

$$(1 + M)^2 - (3M - 1)^2 E^2 = 3M(1 - M)(3M^2 - 2M + 3) > 0.$$

Thus $\eta = P/E > (3M - 1)e$. Now

$$Wq > \eta + e > 3Me, \quad Rp > \eta + e > 3Me,$$

so

$$p, q > 3e.$$

Together with $p + q \leq 1 - 2e$, this gives

$$(6.57) \quad \boxed{3e < p, q < 1 - 5e.}$$

Furthermore

$$(6.58) \quad e < \frac{P}{1+M} \leq \frac{2\sqrt{3}-3}{6} < \frac{1}{12}.$$

For $c = 1 - e$, put $f_c(t) = c^4 - c^2 + ct - t^2$. Concavity and

$$f_c(3e) = e(1 - 7e - 4e^2 + e^3) > 0,$$

$$f_c(1 - 5e) = e(2 - 15e - 4e^2 + e^3) > 0$$

give $f_c(p), f_c(q) > 0$. Since $c > p + q$ and $p + c, q + c > 1$, the exact four support-contact side lengths all exceed one. Hence $c_{\max}(p, q) < 1 - e$.

5.13.4. *Detailed adjacent Vd residual estimate.* In the adjacent placement, the residuals satisfy

$$(6.59) \quad \rho_R + \rho_L < \frac{1}{2}.$$

The signed CE2 domain gives

$$(6.60) \quad \alpha + \delta < \frac{1}{6} \min\{R, W\}.$$

If $\rho_L = W - \alpha$, then

$$4\delta + \alpha \leq 4(\alpha + \delta) < \frac{2W}{3} < W,$$

so $4\delta < \rho_L$.

Otherwise $\rho_L = 1 - A_0$. The opposite residual must be

$$\rho_R = 1 - (W + \delta), \quad B_0 \geq \frac{k}{R}.$$

The boundary-endpoint diameter inequality gives

$$(6.61) \quad \rho_L > \frac{k}{2R} > \frac{\eta}{2R}.$$

Combining this with (6.59) gives

$$\delta > R - \frac{1}{2} + \frac{\eta}{2R} =: J(R).$$

The function J is increasing, $J(3/8) = 1/24$, and (6.60) gives $\delta < 1/24$. Thus $R < 3/8$. Then

$$(2 - 3R)^2 - E^2 = (3 - 8R)(1 - R) > 0,$$

so $\eta/R > 1/3$, and (6.61) yields

$$(6.62) \quad \rho_L > \frac{1}{6} > 4\delta.$$

This proves the residual estimate used by the adjacent Vd terminal. The new one-third radial envelope then gives

$$C_2 < 1 - \frac{\rho_L}{3} < 1 - \delta.$$

5.13.5. *Detailed nonadjacent Vd separation.* Let $x = k/W$ and $y = k/R$ be the two center near endpoints, and let B, U be the farthest already-covered extents. The endpoint-distance inequality and monotonicity of M_0 give

$$(6.63) \quad B \leq M_0(x), \quad U \leq M_0(y).$$

Since

$$1 - M_0(z) > \frac{z}{2} \quad (z > 0),$$

the residual tails satisfy

$$\rho_R > \frac{x}{2}, \quad \rho_L > \frac{y}{2}.$$

The signed center-slack estimate

$$\alpha + \delta < \frac{1}{2} \min\{x, y\}$$

therefore gives

$$(6.64) \quad \alpha + \delta < \min\{\rho_R, \rho_L\}.$$

The Vd own-radial margin then places $\min\{\rho_R, \rho_L\}V_\tau$ beyond the Vd trace, while (6.64) places it beyond the C exit.

5.13.6. *Detailed Vd1 rescue inequalities.* Use the Vd1 corner parameter $t \geq 1$, let $d = \sqrt{t^2 + t + 1}$, and write the supported interval as $[c, u]$. Then

$$c = \frac{t(1-b)}{t+1}, \quad u = \frac{d-a-tb-1}{t}.$$

The midpoint condition gives

$$a + tb \leq d - 1 - \frac{t}{2}.$$

Eliminating $tb = t - c(t+1)$ gives

$$a \leq F(t, c) := d - 1 - \frac{3t}{2} + c(t+1).$$

For fixed $c \leq 1/2$, the function F decreases with t , so

$$a \leq F(1, c) = \sqrt{3} - \frac{5}{2} + 2c =: L(c).$$

The inequality $2L(c) \leq 1 - M_c^{\text{sup}}$ reduces to

$$20c^2 + (18\sqrt{3} - 52)c + 47 - 24\sqrt{3} \geq 0,$$

which holds on $[0, 1/2]$. Hence

$$(6.65) \quad a \leq 1 - M_c^{\text{sup}}.$$

Put $\varepsilon = 1 - u$. Direct substitution gives

$$\varepsilon = \varepsilon_0 + \frac{a}{t}, \quad \varepsilon_0 = \frac{2t+1-c(t+1)-d}{t} > 0.$$

The map

$$z \mapsto \frac{z}{z + \varepsilon_0 + z/t}$$

is increasing, while $\varepsilon_0 + F(t, c)/t = 1/2$. Therefore

$$(6.66) \quad \frac{a}{a + \varepsilon} \leq \frac{F(t, c)}{F(t, c) + 1/2} \leq 2L(c) \leq 1 - M_c^{\text{sup}}.$$

Equations (6.65)–(6.66) are the local hypotheses of the common rescuer-tail lemma.

5.14. Exact direct output formulas. The preceding proofs refer to three local inequalities. We derive them here from Proposition 5.2. This replaces the former presentation by iterated reach maps.

For a nonsupercritical triangle with incoming reach at least a and radial reach at least c , let

$$\mathcal{B}_c(a) = \max\{b : (a, b, c) \text{ is admissible and } a + b \leq 1\}.$$

This is an outgoing capacity for one triangle; it is never composed in the proof.

For $1/2 < c \leq \sqrt{3}/2$, put

$$h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}}.$$

For $\sqrt{3}/2 < c < 1$, put

$$\ell(c) = \frac{c}{2} \left(1 - \sqrt{4c^2 - 3}\right), \quad r(c) = c - \ell(c).$$

Also set

$$Q_-(a, c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a,$$

$$Q_+(a, c) = \frac{2(1 - a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1 - c^2)(1 - a^2)c^2}}.$$

Proposition 5.23 (Four direct high-radial outputs). *For $1/2 < c \leq \sqrt{3}/2$,*

$$(6.66) \quad \mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a, c), & h(c) < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

For $\sqrt{3}/2 < c < 1$,

$$(6.67) \quad \mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a, c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

At $a = \ell(c)$, the selected value is $r(c)$; immediately to the right it is $\ell(c)$.

5.14.1. Low-root and threshold calculations. Let $c = 1 - e$ with $0 < e < 1 - \sqrt{3}/2$. The selected low root $\epsilon(e) = \ell(1 - e)$ satisfies the quadratic

$$(6.68) \quad p_e(z) = z^2 - (1 - e)z + (1 - e)^2 - (1 - e)^4 = 0.$$

Substitution gives

$$p_e(e) = e(1 - 3e + 4e^2 - e^3) > 0.$$

Since $e < (1 - e)/2$, the point e lies below the smaller root. For $U = 2e/(1 - 2e)$,

$$p_e(U) = -\frac{e^2}{(1 - 2e)^2} (4e^4 - 20e^3 + 37e^2 - 28e + 3).$$

The polynomial in parentheses is positive on the required interval, proving

$$e < \epsilon(e) < \frac{2e}{1-2e}.$$

For $e \leq 1/8$,

$$p_e(2e + 5e^2) = e^2(3e + 4)(8e - 1) \leq 0,$$

which proves (6.6).

Now let $A > \epsilon(e)$ and $C \geq 1 - e$. The linear branch of Proposition 5.23 is no longer available. On the selected constant and Q_- pieces, the outgoing reach is at most $\epsilon(e)$. On the Q_+ piece, the selected quadratic has its outgoing coordinate below the same root. Thus $B \leq \epsilon(e)$, proving the direct threshold Lemma 5.3 with no following-edge map.

5.14.2. *The strict-supercritical envelope.* Fix $0 \leq c < 1/2$ and maximize the outgoing boundary reach over strict supercritical pairs. The maximum lies on the S -cell frontier and at the boundary where the incoming coordinate tends to the complementary value. Eliminating that coordinate from $F_S = 0$ gives

$$b^2 - cb + 2c - 1 = 0.$$

The larger selected root is

$$M_c^{\text{sup}} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

The corresponding equality state lies on the boundary $a + b = 1$, so it is not strictly supercritical. Therefore every actual strict-supercritical role has

$$B < M_c^{\text{sup}}.$$

This strict nonattainment is the only supercritical output used in the T3-like and Vd1 rescuer proofs.

5.14.3. *Quarter radial envelope.*

Lemma 5.24 (Quarter radial envelope). *If*

$$0 \leq h_0 \leq p, \quad p + h_0 < 1,$$

then

$$(6.69) \quad \boxed{c_{\max}(p, h_0) \leq 1 - \frac{h_0}{4}}.$$

The inequality is strict when $h_0 > 0$.

5.14.4. *Vd corner margins.* In the exact Vd1/Vd2 corner form, let the boundary reaches be a, b , let $t > 0$, and put $d = \sqrt{t^2 + t + 1}$. The own-radial reach is

$$c_0 = \frac{d - a - tb}{t + 1}.$$

If the role has positive support on the forward adjacent arm, its far endpoint there is

$$u_+ = \frac{d - a - tb - 1}{t}.$$

Since $d < t + 1$,

$$(6.70) \quad c_0 < 1 - \frac{a + tb}{t + 1} \leq 1 - \min\{a, b\},$$

and

$$(6.71) \quad u_+ < 1 - b - \frac{a}{t} < 1 - b.$$

Thus lower boundary demands $a \geq A$, $b \geq H$ imply

$$c_0 < 1 - \min\{A, H\}, \quad u_+ < 1 - H.$$

Reflection supplies the backward adjacent-arm statements. These two one-line inequalities are exactly the radial separations used in the one-Vd placement proof.

5.15. Casewise construction of the forced finite sets. This subsection records the exact forced object used by each nonzero-gap row.

5.15.1. *One gap and no supercritical V role.* Normalize the gap to

$$J = [B_0, 1 - A_1] = [\ell, r] \subset e_{0,1},$$

and put $p = 1 - r$, $q = \ell$. Handoffs on the other five edges and nonsupercriticality show that every local pair dominates (p, q) . Let $c_* = c_{\max}(p, q)$. Type-aware radial forcing gives

$$(6.72) \quad (1 - c_*)V_i \in U_C \quad (0 \leq i \leq 5).$$

Their convex hull contains $\mathcal{D}_{h(1-c_*)}$, while the actual gap is exactly $J(p, q)$. This is the complementary-gap terminal.

The same construction works with one or two T3-like roles because the permitted neighboring capacity is bounded by c_* .

5.15.2. *Two gaps.* For CE2, put

$$p = W - \alpha, \quad q = R - \delta.$$

The four center-free handoffs between the two gap edges show that $T_1, \dots, T_5$ dominate (p, q) . Therefore

$$(6.73) \quad D_2 = (1 - c_{\max}(p, q))V_2, \quad D_4 = (1 - c_{\max}(p, q))V_4$$

are missed by all V roles. The shortened CE2 theorem proves that one lies beyond the corresponding C exit. This argument is independent of the criticality of T_0 and permits the type-aware T3-like neighboring terms.

5.15.3. *One gap, one supercritical Vd0 role.* Let C_i be the actual own-radial reach and $P_i = (1 - C_i)V_i$. Every P_i is center-forced, but the enclosure certificate retains only

$$(6.74) \quad \boxed{K_{\text{tr}} = \{O, M_0, X(\ell), X(r), P_2, P_3, P_4\}.}$$

Suppose an open unit triangle U' contains this set. Its actual gap trace gives, with

$$Q = \frac{\eta + \alpha + \delta}{2R},$$

the endpoint-diameter upper squeeze

$$A_0 < 1 - Q.$$

Containment of P_4, P_3, P_2 supplies the three radial demands used by the CE1 transverse return. In CE2, P_4, P_2 supply the two threshold demands. Both center classes force

$$A_0 > 1 - Q.$$

Thus $\Lambda(K_{\text{tr}}) \geq 1$. The former points P_0, P_1, P_5 are not needed.

5.15.4. *The T3-like rescuer endpoint.* Write the T3-like supported interval on r_1 as $[c, u]$. The point

$$P_T = (1 - u)V_1$$

is missed by the open T3-like role, the adjacent supercritical role, and all Vd0 or nonlocal roles. Thus $P_T \in U_C$.

The T3 normal form proves

$$a \leq 1 - M_c^{\sup}, \quad \frac{a}{a + 1 - u} \leq 1 - M_c^{\sup}, \quad a + 1 - u \leq 1.$$

These are precisely the hypotheses of the common rescuer-tail budget. No T3-specific center-hiding calculation remains.

5.15.5. *The Vd radial points.* In the adjacent placement, the Vd supported endpoint lies below $1 - \rho_L$. The ordinary role T_2 has boundary demands

$$\left(\frac{1}{2} + \rho_R, \rho_L \right).$$

The one-third radial envelope gives

$$C_2 < 1 - \frac{\rho_L}{3} < 1 - \delta.$$

Therefore a point of r_2 between the two local V traces and the C interval is uncovered.

In the nonadjacent placement,

$$D_\tau = \min\{\rho_R, \rho_L\}V_\tau$$

lies beyond the Vd own trace and beyond every relevant C exit.

In the Vd1 neighboring-midpoint placement, the O-side supported endpoint is center-forced and the two local inequalities are routed to the same rescuer-tail budget used by the T3-like case.

5.15.6. *The corrected replacement placement.* When neither distinguished role is based at V_0 , the corrected Vd1 two-chart replacement preserves the covered skeleton and produces six nonsupercritical Vd0 roles. Recompute the output gap rank. Rank zero uses the boundary-complete length theorem, rank one the common radial disk and actual gap, and rank two the shortened CE2 theorem. No preservation of the input gap rank is needed.

5.16. **Endpoint, openness, and selector audit.** The arguments above repeatedly pass between open role traces, closed maximal reaches, and compact witness sets. We record the precise endpoint statements used in those passages.

Lemma 5.25 (Open trace endpoint). *Let U be an open equilateral triangle and $T = \bar{U}$. If $T \cap [V, O]$ is the closed interval from V to a point P , then $P \notin U$. If Q is a noncentral point of a radial arm and $Q \in U$, then U has positive-length intersection with that arm.*

Lemma 5.26 (Compact set in an open unit triangle). *If a compact set K is contained in an open unit equilateral triangle, then*

$$\Lambda(K) < 1.$$

Lemma 5.27 (Continuity of the enclosure number). *For nonempty compact sets K, L ,*

$$|\Lambda(K) - \Lambda(L)| \leq \frac{3}{h} d_H(K, L),$$

where d_H is the Hausdorff distance.

These lemmas justify all singleton-gap limits and every use of a radial trace endpoint as a center-forced point.

5.16.1. *Branchwise proof of common-pair domination.* We expand the neighboring-capacity comparison in Lemma 5.10. Assume first that $q \leq p$. The own-radial demand $1 - q$ is admissible because the exact edge-normal orientation has side

$$q + \max\{p, 1 - q\} = 1.$$

Thus

$$(6.73a) \quad c_{\max}(p, q) \geq 1 - q.$$

For $C_+(p, q)$ there are three possibilities.

On the linear branch,

$$C_+(p, q) = 1 - q.$$

On the plateau branch, $C_+ = p(p)$, where the cubic root satisfies

$$f_p(1 - q) = (1 - q)^3 - (p + 2)(1 - q)^2 + 2(p + 1)(1 - q) - 1 \geq 0$$

throughout the selected interval. Since the cubic is increasing, $p(p) \leq 1 - q$. On the radical branch, the capacity decreases from its selected transition value and therefore is no larger than the plateau. Hence

$$C_+(p, q) \leq 1 - q.$$

The reflected calculation gives the same bound for C_- and treats the order $p \leq q$. Combining with (6.73a) proves

$$C_{\pm}(p, q) \leq c_{\max}(p, q).$$

The same argument is valid for any actual pair (a, b) with $a \geq p$ and $b \geq q$, because the family of triangles containing the larger anchors is a subfamily of the one containing the smaller anchors. Thus the relevant capacity is coordinatewise nonincreasing.

5.16.2. *Finite-caliper selector audit.* For a finite point set K , Proposition 5.1 reduces $\Lambda(K)$ to finitely many orientations. We spell out the reduction used for every diagram and for the anisotropic one-gap set. Let the convex hull vertices be $P_0, \dots, P_{r-1}$ in cyclic order. On an angular interval on which the three support points are fixed, write

$$H(\theta) = \sum_{j=0}^2 \langle Q_j, n_{\theta+2\pi j/3} \rangle.$$

Then

$$H''(\theta) = -H(\theta).$$

The support sum is positive unless K is a singleton at the origin. Hence an interior critical point is a strict maximum. A minimum occurs when two support points

tie in one normal direction. That normal is perpendicular to a hull edge $P_i P_{i+1}$. Therefore

$$(6.73b) \quad \Lambda(K) = \min_{0 \leq i < r} \frac{1}{h} \sum_{j=0}^2 h_K(R^j n_i),$$

where n_i is either unit normal to $P_i P_{i+1}$.

This finite-caliper identity has two consequences. First, the enclosure number is unchanged by replacing a witness set by its convex hull. Second, no numerical angular minimization is hidden in the proof: every terminal can be checked at finitely many support contacts. The complementary-gap theorem uses this fact analytically; the zero-gap nine-point theorem uses it through the support-arc certificate in the final appendix.

5.16.3. *Terminal upper bound in the one-gap exact-one case.* We give the elementary derivation of (6.38). The active center trace begins at k/R , so the unique supercritical role has forward boundary reach

$$B_0 \geq \frac{k}{R}.$$

The center exit on r_0 is $1 - k$. Since the finite set contains the actual radial endpoint $P_0 = (1 - C_0)V_0$, containment by the candidate center gives

$$C_0 \geq k.$$

The closure of T_0 therefore contains three points based at V_0 with selected reaches

$$A_0, \quad \frac{k}{R}, \quad k.$$

Discarding the radial point only weakens the diameter condition, and gives

$$A_0 \leq M_0\left(\frac{k}{R}\right).$$

For $0 < z < 1$,

$$M_0(z) < 1 - \frac{z}{2}.$$

Indeed, after moving terms and squaring, this is equivalent to $3z^2 > 0$. With $z = k/R$,

$$A_0 < 1 - \frac{k}{2R} = 1 - Q.$$

This is the terminal capacity contradicted separately by the direct CE1 and CE2 reach arguments.

5.16.4. *Strictness at singleton gaps.* Suppose the V-gap on $e_{0,1}$ is the singleton $X(\ell)$. Although its length is zero, neither incident open V trace contains the point. Boundary locality therefore forces $X(\ell) \in U_C$. Choose $r_n > \ell$ decreasing to ℓ , and reduce the incoming reach at T_1 from $1 - \ell$ to $1 - r_n$. Every local diameter inequality remains feasible, and the strict supercritical pattern is unchanged for large n . The non-singleton-gap witness sets converge to the singleton witness set. By Lemma 5.27, the inequality $\Lambda \geq 1$ passes to the limit. Thus no proof in this section charges numerical gap length; every one uses the actual missing point.

5.16.5. *Why a common disk is insufficient for $N_+ = 1$.* The one-gap $N_+ = 1$ finite set retains all six actual radial endpoints rather than replacing them by one common radius. This is necessary. The boundary sequence has two ascents when the gap jump is included: the V-gap jump and the unique supercritical V triangle. They need not join the global minimum to the global maximum. A common lower pair can therefore move several radial witnesses toward O and can produce a disk-plus-gap set with enclosure number below one. The anisotropic set $P_0, \dots, P_5$ preserves the six actual radial constraints, and equations (6.76), (6.38), and the direct CE1/CE2 arguments use those constraints individually. This is the precise point at which the new proof differs from a relabeling of the former propagation calculation.

Proposition 5.28 (All nonzero-gap finite-enclosure rows). *Every nonzero-gap row of Table 1 assigned to Strategy 3 is impossible.*

5.17. **The explicit zero-gap nine-point obstruction.** The explicit zero-gap member of the finite-enclosure strategy applies when all six V triangles are Vd0, the boundary is covered, and exactly one V triangle is supercritical. Six radial points and three asymmetric frontier points are forced into the center triangle. The detailed construction is followed by the support-arc covering argument and the final enclosure contradiction.

5.17.1. *Detailed Nine-Point Enclosure Argument.* This section supplies the exact frontier, witness, sign, and cap-overlap calculations for the direct nine-point obstruction.

This section proves a center-class-independent theorem. Its hypotheses are that all six V triangles are Vd0, that they cover the full boundary, and that exactly one actual boundary V triangle is supercritical. It therefore applies to every $N_{\text{gap}} = 0$ case with these V triangle data, independently of whether the center is CE0, CE1, or CE2. Six common radial points and three asymmetric points are excluded directly from all six actual V triangles and are therefore forced into the C triangle. Only the unique supercritical V triangle uses an AB -union; no nonsupercritical model core is introduced.

The finite caliper criterion. We first state the finite caliper criterion used below. It enters the proof through the exact statement and verification argument given here; neither a plot nor an unrecorded numerical convention is used.

Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For a nonempty finite set $S \subset \mathbb{R}^2$ and a vector N , put

$$(30) \quad W_S(N) = \sum_{k=0}^2 \max_{z \in S} \langle z, R^k N \rangle.$$

Theorem 5.29 (Finite caliper certificate). *If S has at least two distinct points, then S is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in S$ and $\varepsilon \in \{-1, 1\}$ for which*

$$(31) \quad W_S(\varepsilon J(q - p)) \leq \frac{\sqrt{3}}{2} \|p - q\|.$$

If S consists of one point, it is contained in equilateral triangles of arbitrarily small positive side length.

For $S_x = \{O, A, B, x\}$, every clause of (31) is a finite polynomial clause of degree at most four in the coordinates of A, B, x ; at most $12 \cdot 4^3$ clauses are required, and repeated point labels merely delete zero-length pairs.

Corner-clipped AB -unions. Let e_A, e_B be unit vectors with angle 120° , put

$$O = 0, \quad A = ae_A, \quad B = be_B, \quad C = \{ue_A + ve_B : u, v \geq 0\},$$

and define

$$(32) \quad \mathcal{R}(a, b) = C \cap \bigcup \{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}.$$

The finite description in Theorem 5.29 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (32) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem 5.30 (Strict supercritical AB -union). *Assume*

$$a + b > 1, \quad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\begin{aligned} \alpha &= \frac{h(a + 2b - aD)}{2\rho}, & \beta &= \frac{h(a - b + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-a + b + (a + b)D)}{2\rho}, & \delta &= \frac{h(2a + b - bD)}{2\rho}. \end{aligned}$$

Then all four coefficients are positive and

$$(33) \quad \mathcal{R}(a, b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B),$$

where

$$\begin{aligned} \mathcal{D}_A &= \{(u, v) : (u - a)^2 + v^2 - (u - a)v \leq 1\}, \\ \mathcal{D}_B &= \{(u, v) : u^2 + (v - b)^2 - u(v - b) \leq 1\}, \\ \mathcal{H}_A &= \{(u, v) : \alpha(u - a) + \beta v \leq 0\}, \\ \mathcal{H}_B &= \{(u, v) : \gamma u + \delta(v - b) \leq 0\}. \end{aligned}$$

Its non-axis frontier consists, in counterclockwise order, of the A -circle, the A -line, the B -line, and the B -circle.

Exact-one handoffs and common reach bounds.

Lemma 5.31 (Exact-one handoff chain). *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.10 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \quad b = \xi_4, \quad p = 1 - b, \quad q = 1 - a,$$

one has

$$(34) \quad 0 < a, b < 1, \quad a + b > 1, \quad a^2 + ab + b^2 < 1,$$

and every selected V triangle obeys

$$a_i \geq p, \quad b_i \geq q.$$

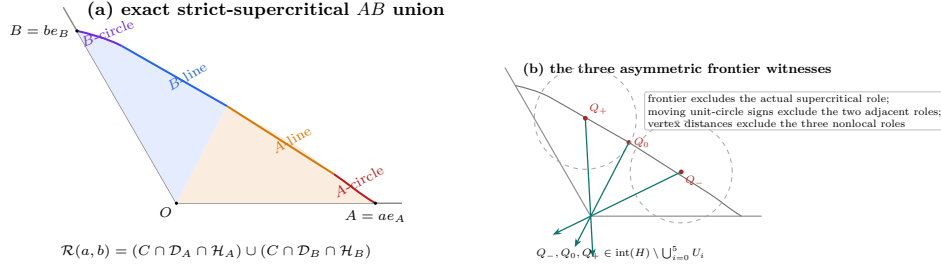


FIGURE 20. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an A -circle arc, an A -line segment, a B -line segment, and a B -circle arc. The points Q_-, Q_0, Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

Six directly forced radial points. Put $h = \sqrt{3}/2$ and

$$s = p + q, \quad m = \min(p, q), \quad M = \max(p, q), \quad \chi = s^4 - s^2 + pq.$$

The strict domain (34) gives

$$(35) \quad pq > (1-s)(2-s), \quad m > 1-s, \quad s > 1-m.$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition 5.2. On the present subcritical domain, equation (6.2) becomes

$$(36) \quad c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s .

Lemma 5.32 (Direct radial forcing). *One has*

$$(37) \quad 0 < c_* < 1, \quad c_* > 1 - m.$$

Assume that every U_i is Vd0 and that $U_C, U_0, \dots, U_5$ cover H . Then, for every i ,

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$(38) \quad \mathcal{D}_\eta = \{x : \|x\| \leq \eta\}, \quad \eta = h(1 - c_*).$$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \quad \mathbf{q}(u, v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a . Accordingly, set

$$\begin{aligned}\alpha &= \frac{h(b+2a-bD)}{2\rho}, & \beta &= \frac{h(b-a+(a+b)D)}{2\rho}, \\ \gamma &= \frac{h(-b+a+(a+b)D)}{2\rho}, & \delta &= \frac{h(2b+a-aD)}{2\rho},\end{aligned}$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of Theorem 5.30 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \quad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$(39) \quad J = \left(\frac{2b+a-aD}{D+3}, \frac{b+2a-bD}{D+3} \right).$$

Put

$$E_- = (1-b, 2-b), \quad E_+ = (2-a, 1-a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1+t, t), \quad 1-a \leq t \leq b, \quad \text{and} \quad Z_3(y) = (y, 1+y), \quad 1-b \leq y \leq a.$$

The restrictions of the two unit-circle equations are

$$(40) \quad \begin{aligned}g_-(\lambda) &= \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2, \\ g_+(\mu) &= \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2.\end{aligned}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\begin{aligned}\lambda_- &= 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}.\end{aligned}$$

Lemma 5.33 (Moving adjacent-triangle signs). *Throughout (34),*

$$(41) \quad \lambda_* < \lambda_- < h^{-1}, \quad \mu_* < \mu_+ < h^{-1}.$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P, t) = \mathbf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1-a, b]$,

$$(42) \quad F(L_+(\mu_+), t) \geq 0, \quad F(J, t) > 0, \quad F(L_-(\lambda_-), t) > 0.$$

Under the reflection $(u, v, a, b) \leftrightarrow (v, u, b, a)$, the analogous signs hold for every $y \in [1-b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Define the local and global witnesses by

$$(43) \quad Q_-^{\text{loc}} = L_-(\lambda_-), \quad Q_0^{\text{loc}} = J, \quad Q_+^{\text{loc}} = L_+(\mu_+),$$

and by the displayed conversion from (u, v) to the plane.

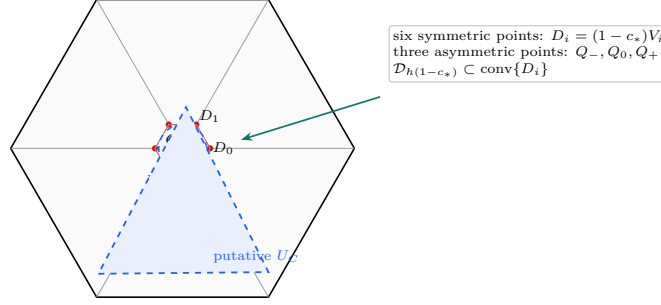
Lemma 5.34 (Direct asymmetric forcing). *Assume that $U_C, U_0, \dots, U_5$ cover H . Then*

$$Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i,$$

and consequently all three witnesses belong to U_C .

The nine directly forced points are $D_0, \dots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas 5.32 and 5.34 therefore force the compact set

$$(44) \quad K_{\text{wit}}(a, b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$



$K_{\text{wit}} = \mathcal{D}_{h(1-c_*)} \cup \{Q_-, Q_0, Q_+\} \subset U_C$ under a hypothetical cover

FIGURE 21. The zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB -frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

Tangent reduction and exact mixed overlaps. For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (40), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g'_-(\xi_*)}, \quad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g'_+(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_- \right), \\ B &= J, \\ C &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+ \right), \end{aligned}$$

and convert them to global vectors.

Lemma 5.35 (Newton inner reduction). *The points above have coordinates rational in a, b, D and satisfy*

$$A \in (Q_0, Q_-), \quad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \text{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

obeys $\widehat{K} \subseteq \text{conv}(K_{\text{wit}})$.

Put $W = C + RA$. For a disk $\mathbb{D}(X, r)$ let

$$I(X, r) = \{n : \|n\| = 1, \langle X, n \rangle + r \geq h\}.$$

Ray order and four overlaps.

Proposition 5.36 (Technical four-overlap verification). *Assume (34) and $c_* > 2/3$. Then*

$$\|A\|, \|C\| > \eta,$$

the rays A, B, C, W, RA occur in that cyclic order, and the four consecutive cap pairs

$$I(A, 2\eta) \leftrightarrow I(B, 2\eta) \leftrightarrow I(C, 2\eta) \leftrightarrow I(W, \eta) \leftrightarrow I(RA, 2\eta)$$

overlap across their intervening short sectors.

5.17.2. *Support arcs and the enclosure conclusion.* Let R denote counterclockwise rotation through $2\pi/3$. Retain the points A, B, C , the vector $W = C + RA$, the forced disk $\mathcal{D}_\eta$, and the exact witness set K_{wit} constructed above. For uniform notation put

$$P_- = A, \quad P_0 = B, \quad P_+ = C, \quad P_* = W, \quad r_* = \eta,$$

and

$$\widehat{K} = \text{conv}(\mathcal{D}_\eta \cup \{A, B, C\}).$$

For $X \in \mathbb{R}^2$ and $r \geq 0$, write

$$\mathbb{B}(X, r) = \{Y : \|Y - X\| \leq r\},$$

and define the level- h , $h = \sqrt{3}/2$, support arc

$$\text{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

Lemma 5.37 (Cyclic support-arc covering). *Let K be compact and put*

$$\Sigma_K = K + RK + R^2K.$$

Suppose Σ_K contains the full R -orbits of

$$\mathbb{B}(P_-, 2r_*), \quad \mathbb{B}(P_0, 2r_*), \quad \mathbb{B}(P_+, 2r_*), \quad \mathbb{B}(P_*, r_*).$$

Assume that the rays

$$P_-, P_0, P_+, P_*, RP_-$$

occur in this cyclic order, that every corresponding support arc is connected with half-width less than $\pi/2$, and that each consecutive pair of arcs intersects across the intervening short angular sector. Then

$$\Lambda(K) \geq 1.$$

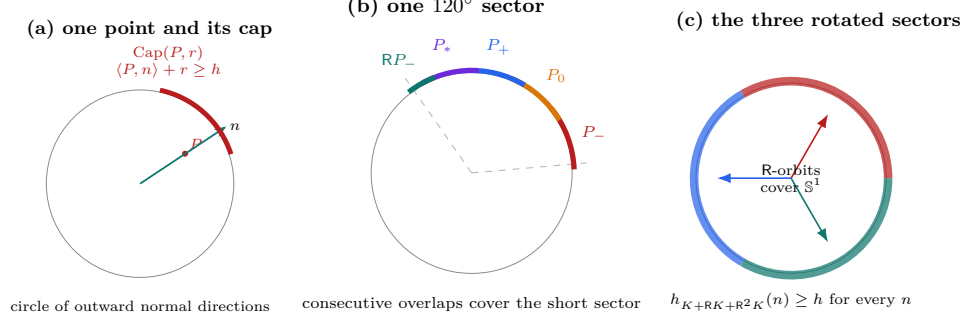


FIGURE 22. The support-cap conclusion for the zero-gap witness. A point and a disk radius determine an arc of normal directions in which their support is at least h . The four consecutive overlap checks cover the sector from the ray of P_- to that of RP_- , and the three 120° rotations cover the full normal circle. The arc widths are schematic; the cyclic order and overlap pattern are the exact hypotheses of the support-arc lemma.

Theorem 5.38 (Witness enclosure). *For every exact-one-supercritical parameter pair (a, b) constructed above,*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Theorem 5.39 (Zero-gap obstruction). *There is no cover of H by open unit equilateral triangles $U_C, U_0, \dots, U_5$ such that every U_i is Vd0, the six vertex triangles cover ∂H , and exactly one V triangle is supercritical.*

Proposition 5.40 (Cases excluded by the nine-point obstruction). *Theorem 5.39 excludes every $N_{\text{gap}} = 0$, $N_+ = 1$, all-Vd0 case, independently of the center class.*

6. COMPLETION OF THE PROOF

Corollary 1.2 follows from Proposition 2.1.

APPENDIX A. EXACT UNEQUAL-RADIUS SUPPORT-ARC INTERSECTIONS

The two mixed cap overlaps in Method 4 are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms; the repository provenance manifests identify the accompanying code and sparse integer data. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

A.1. Certificate manifest. The machine-readable provenance manifest identifies the six sparse-data shards and both exact verifiers by full Git blob; the source ledger records their role. They decode to one canonical sparse-polynomial transcript with SHA-256 digest

dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

A.2. Rational radial upper envelopes. Retain the notation of Section 5.17.1. Thus

$$p = 1 - b, \quad q = 1 - a, \quad s = p + q, \quad m = \min\{p, q\}, \quad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \quad h = \frac{\sqrt{3}}{2}, \quad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$(45) \quad mM > (1 - s)(2 - s), \quad m > 1 - s, \quad s > \frac{2}{3}.$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \quad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$(46) \quad \bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \leq 0), \quad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \geq 0).$$

Lemma A.1 (Branchwise radial upper bounds). *On the corresponding radial cells,*

$$c_* \leq \bar{c}_L < 1, \quad c_* \leq \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_$.*

Reflect so that $z = a - b \geq 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \quad m = \frac{1 - U - z}{2}, \quad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

and

$$\Xi(U, z) = 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2,$$

$$G(U, z) = 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z.$$

Thus

$$(47) \quad \bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \quad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}.$$

A.3. Gram reduction and the geometric implication. On the active cell let $\bar{c}$ denote the corresponding expression in (47), and put

$$\bar{\eta} = h(1 - \bar{c}), \quad e = 1 - \bar{c}, \quad t = 2\bar{c} - 1.$$

Lemma A.1 gives $0 < \bar{\eta} \leq \eta = h(1 - c_*)$. It therefore suffices to prove the mixed intersections for the smaller disks with radii $2\bar{\eta}, \bar{\eta}, 2\bar{\eta}$.

Represent a global vector as (x, hy) and put $C' = R^{-1}C$. Define

$$N_A = \|A\|^2, \quad N_C = \|C'\|^2, \quad \nu = \langle A, C' \rangle, \\ \Delta_0 = x_A y_{C'} - y_A x_{C'} > 0.$$

Thus the Euclidean cross product is $h\Delta_0$, and the Gram identity is

$$(48) \quad h^2 \Delta_0^2 = N_A N_C - \nu^2.$$

Define

$$(49) \quad Q_A = \Delta_0^2 - \|tA - eC'\|^2, \quad Q_C = \Delta_0^2 - \|tC' - eA\|^2.$$

Lemma A.2 (Residual signs imply the mixed cap intersections). *If $Q_A, Q_C \geq 0$, then*

$$I(C, 2\bar{\eta}) \cap I(C + RA, \bar{\eta}) \neq \emptyset,$$

and

$$I(C + RA, \bar{\eta}) \cap I(RA, 2\bar{\eta}) \neq \emptyset.$$

Consequently the same intersections hold with η in place of $\bar{\eta}$.

A.4. Eight exact integer-polynomial signs. The Newton witnesses A, C are rational functions of U, z, D . Substituting (47) into (49), clearing denominators, and reducing by

$$D^2 - K(U, z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$(50) \quad \text{num}(Q_{B,X}) = 32(K+3)^2 \{A_{B,X}(U, z) + DB_{B,X}(U, z)\},$$

where all eight core polynomials lie in $\mathbb{Z}[U, z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U, z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C , both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

A.5. Three global positive-basis certificates. Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \quad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_\Delta = 1 - \frac{\sqrt{3}}{2}, \quad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

$$\begin{aligned} T : & \quad 0 \leq U \leq U_\Delta, \quad 0 \leq z^2 \leq \mathcal{P}(U), \\ L_0 : & \quad 0 \leq U \leq U_\Delta, \quad \mathcal{P}(U) \leq z^2 \leq \mathcal{G}(U), \\ L_1 : & \quad U_\Delta \leq U \leq U_{\max}, \quad 0 \leq z^2 \leq \mathcal{G}(U). \end{aligned}$$

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \quad z = y \frac{9-x^4}{8x^2}, \quad 1 \leq x \leq \sqrt{3}, \quad 0 \leq y \leq 1,$$

and then $x = 1 + (\sqrt{3} - 1)r$. This maps $[0, 1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x , the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88, 22)$	2047	2
$B_{T,A}$	$(80, 20)$	1701	0
$A_{T,C}$	$(88, 22)$	2047	2
$B_{T,C}$	$(80, 20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L -cell polynomial write

$$F(U, z) = E_F(U, w) + zO_F(U, w), \quad w = z^2,$$

and

$$R_F(U, w) = E_F(U, w)^2 - wO_F(U, w)^2.$$

The implications $E_F \geq 0$ and $R_F \geq 0$ give $E_F \geq \sqrt{w}|O_F|$, hence $F \geq 0$ for $z \geq 0$. Use the exact charts

$$U = U_\Delta r, \quad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for L_0 , and

$$U = U_\Delta + (U_{\max} - U_\Delta)r, \quad w = s\mathcal{G}(U)$$

for L_1 . The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r, s) = \sum a_{ij} r^i s^j$ has bidegree (m, n) , its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \leq k} \sum_{j \leq \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem A.3 (Exact mixed-overlap certificate). *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in (50) are nonnegative. Hence $Q_A, Q_C \geq 0$, and the two mixed cap intersections required by Proposition 5.36 hold.*

Remark A.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

```
verify_mixed_overlap_core_derivation.py
verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.

APPENDIX B. COMPLETE TRACE-EXACT GEOMETRIC ATLAS

This appendix collects the complete sampled atlas for the finite-enclosure case table. Each shaded region is a dense numerical guide to a source-conditioned trace-exact envelope; it is not used as a proof object. The exact objects are the source-family unions defined in Subsection 5.2, and the proofs use only the set inclusions and endpoint identities stated there.

B.1. Trace-exact AB-envelope atlas. Figures 23–30 show the normalized subcases in Table 2. Each shaded trace-exact envelope is a dense union of valid source triangles; the exact proof object is the source-conditioned union in Subsection 5.2. The dashed equilateral triangle is the numerical minimum for the displayed finite witness set and is included only to explain the terminal geometry.

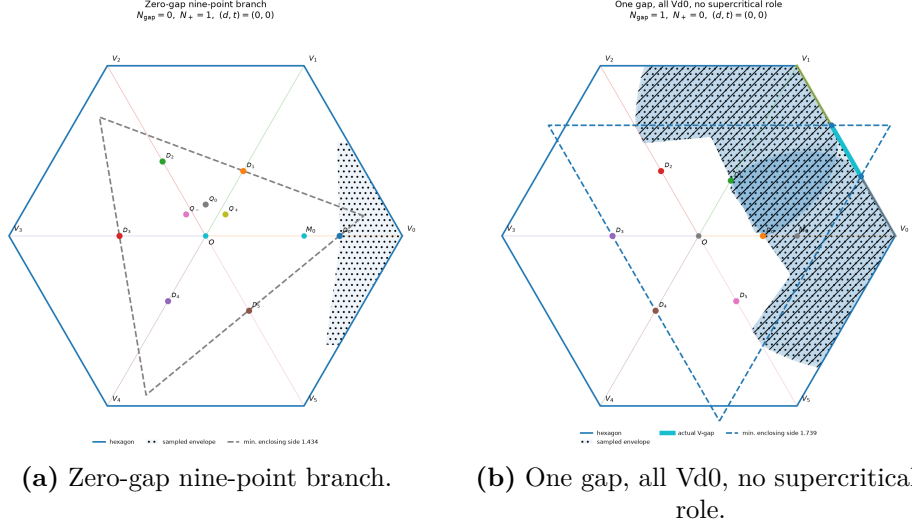


FIGURE 23. Trace-exact AB-envelope snapshots: Zero-gap nine-point branch; One gap, all Vd0, no supercritical role.

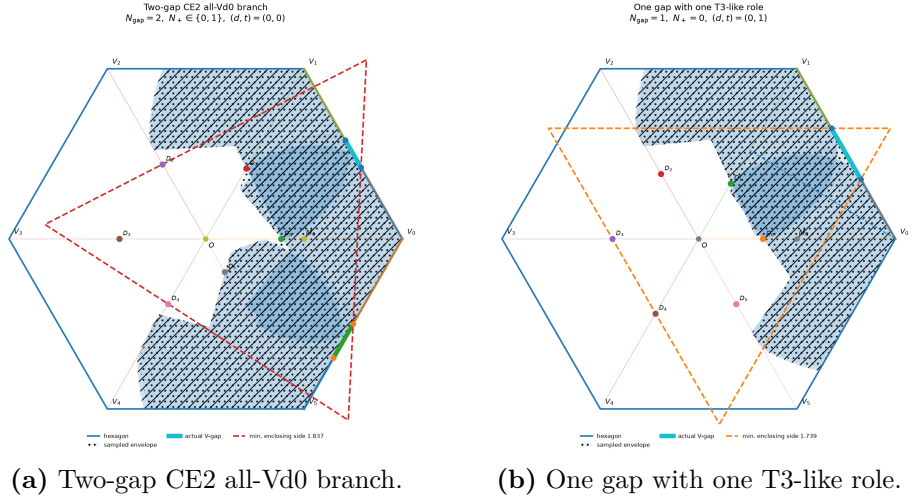


FIGURE 24. Trace-exact AB-envelope snapshots: Two-gap CE2 all-Vd0 branch; One gap with one T3-like role.

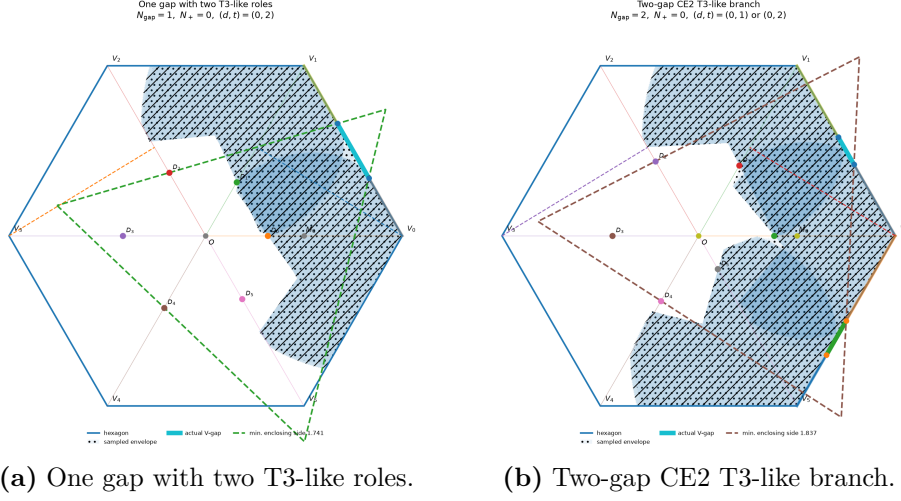


FIGURE 25. Trace-exact AB-envelope snapshots: One gap with two T3-like roles; Two-gap CE2 T3-like branch.

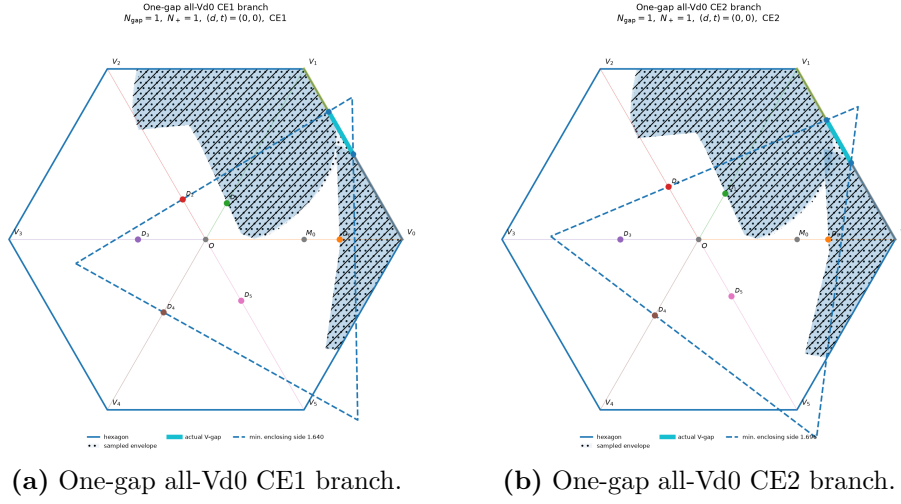
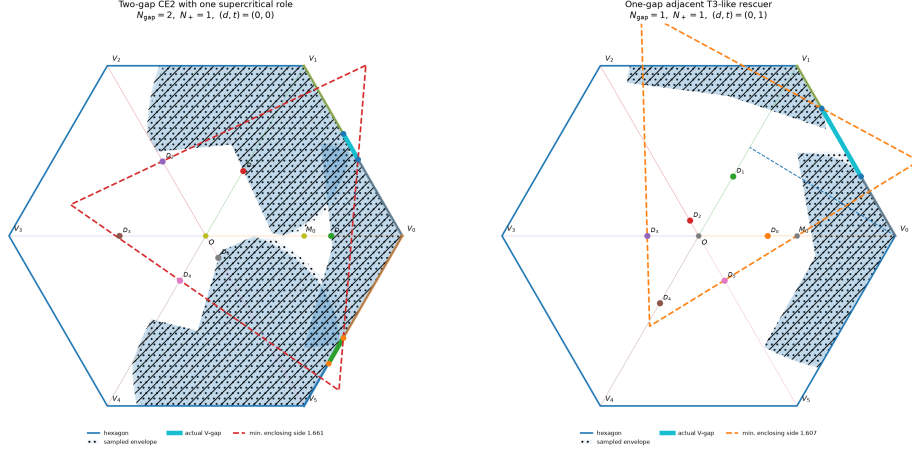
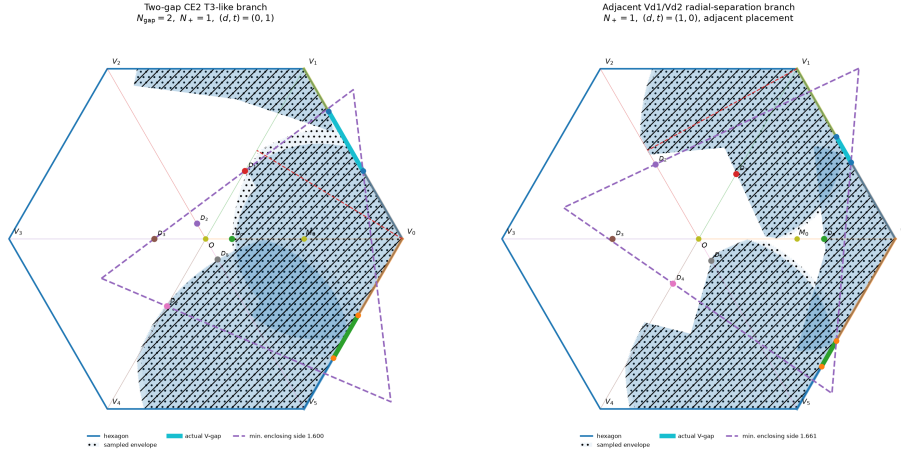


FIGURE 26. Trace-exact AB-envelope snapshots: One-gap all-Vd0 CE1 branch; One-gap all-Vd0 CE2 branch.



(a) Two-gap CE2 with one supercritical role. (b) One-gap adjacent T3-like rescuer.

FIGURE 27. Trace-exact AB-envelope snapshots: Two-gap CE2 with one supercritical role; One-gap adjacent T3-like rescuer.



(a) Two-gap CE2 T3-like branch. (b) Adjacent Vd1/Vd2 radial-separation branch.

FIGURE 28. Trace-exact AB-envelope snapshots: Two-gap CE2 T3-like branch; Adjacent Vd1/Vd2 radial-separation branch.

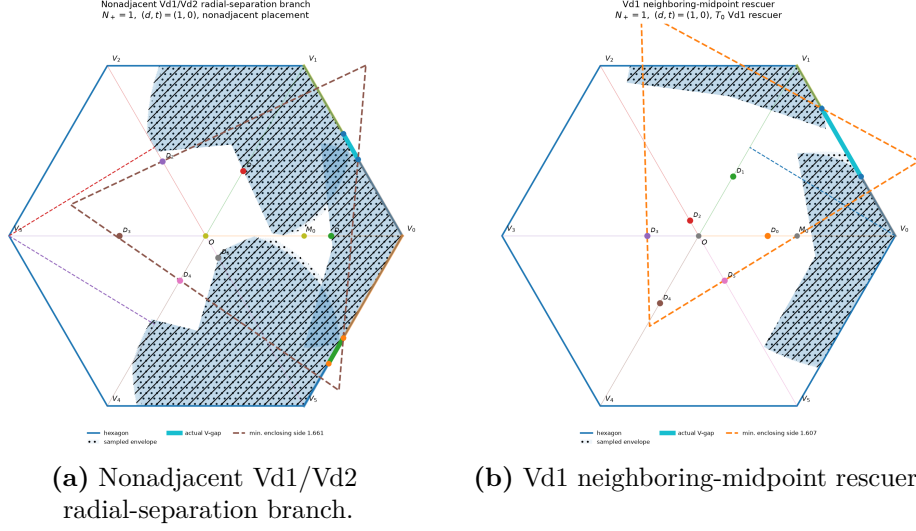


FIGURE 29. Trace-exact AB-envelope snapshots: Nonadjacent Vd1/Vd2 radial-separation branch; Vd1 neighboring-midpoint rescuer.

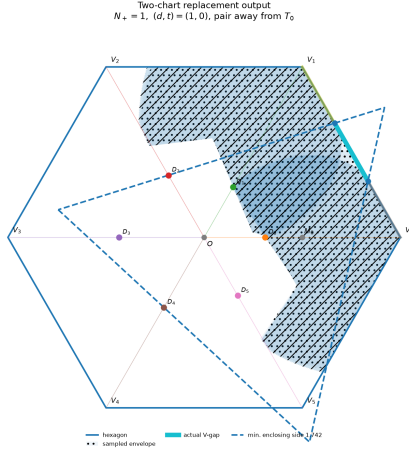


FIGURE 30. Trace-exact AB-envelope snapshots: Two-chart replacement output.